

AI-VALIDATED URBAN GOVERNANCE

UrbanFix

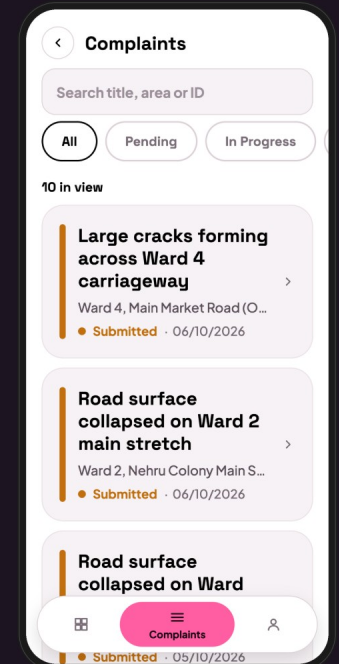
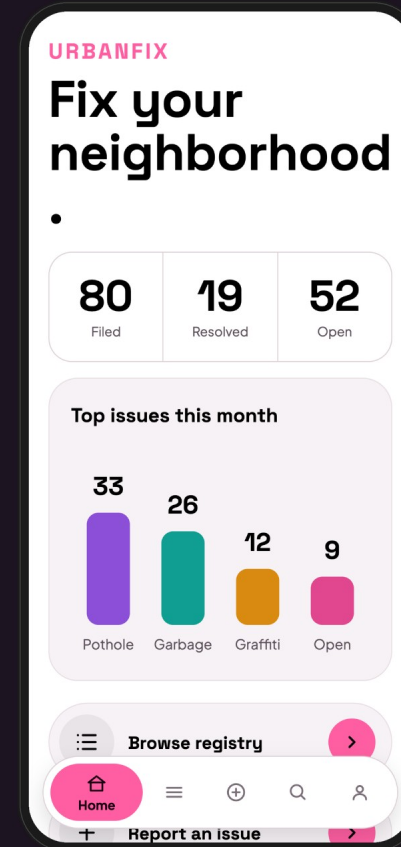
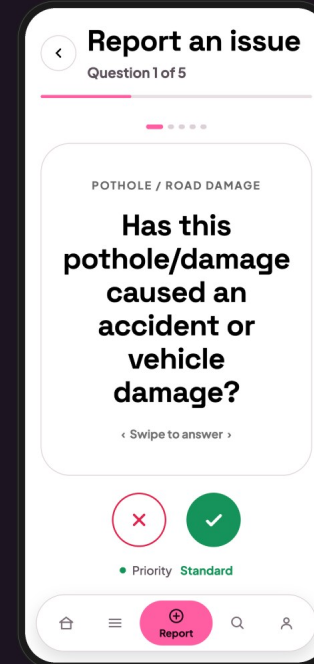
Report

Validate

Act

Prove

Learn

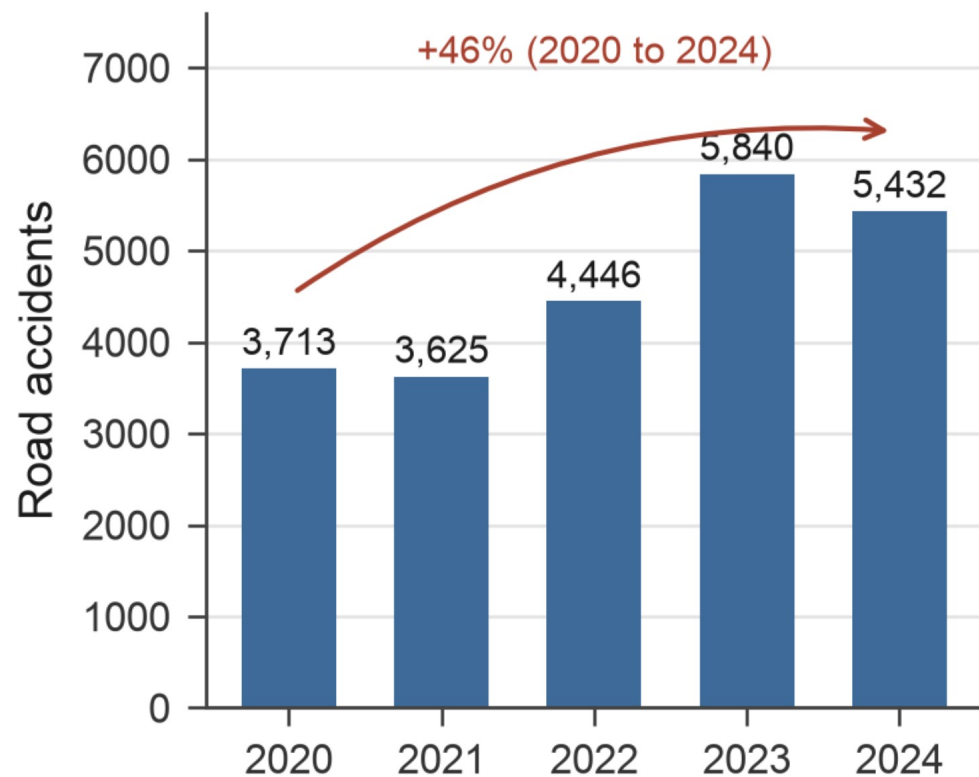


01

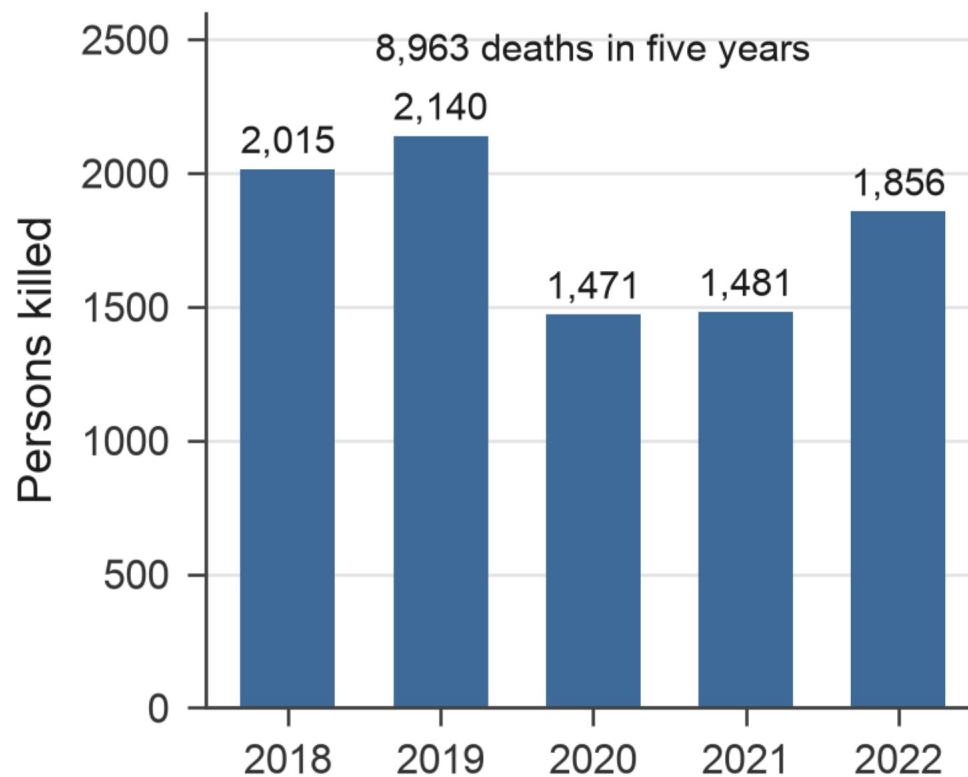
Problem

POTHOLE ACCIDENTS IN INDIA

(a) Accidents attributed to potholes

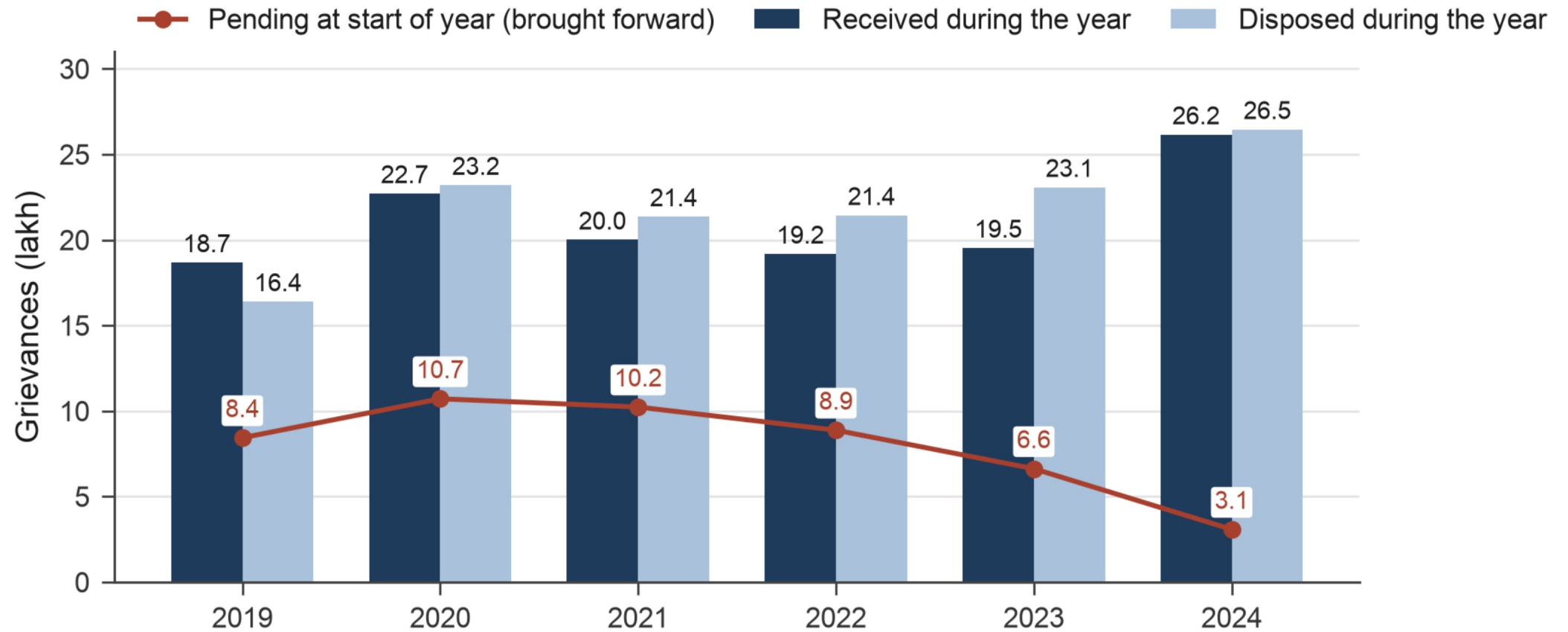


(b) Deaths in pothole-related accidents



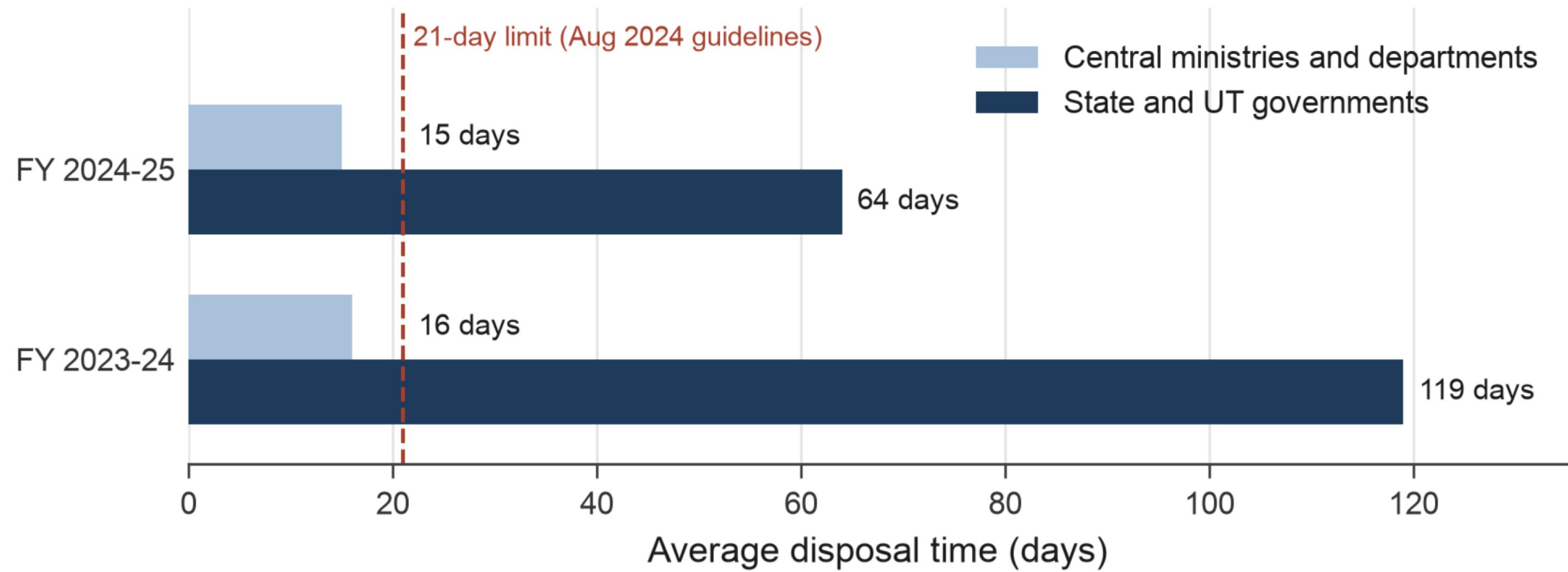
Sources: (a) MoRTH data laid in the Lok Sabha, 12 Feb 2026. (b) MoRTH, Road Accidents in India 2018 to 2022.

GRIEVANCE BACKLOG



Source: DARPG, CPGRAMS figures in Parliament replies (PIB, 16 Mar 2022 and 2 Apr 2025). 1 lakh = 100,000.

DISPOSAL TIME

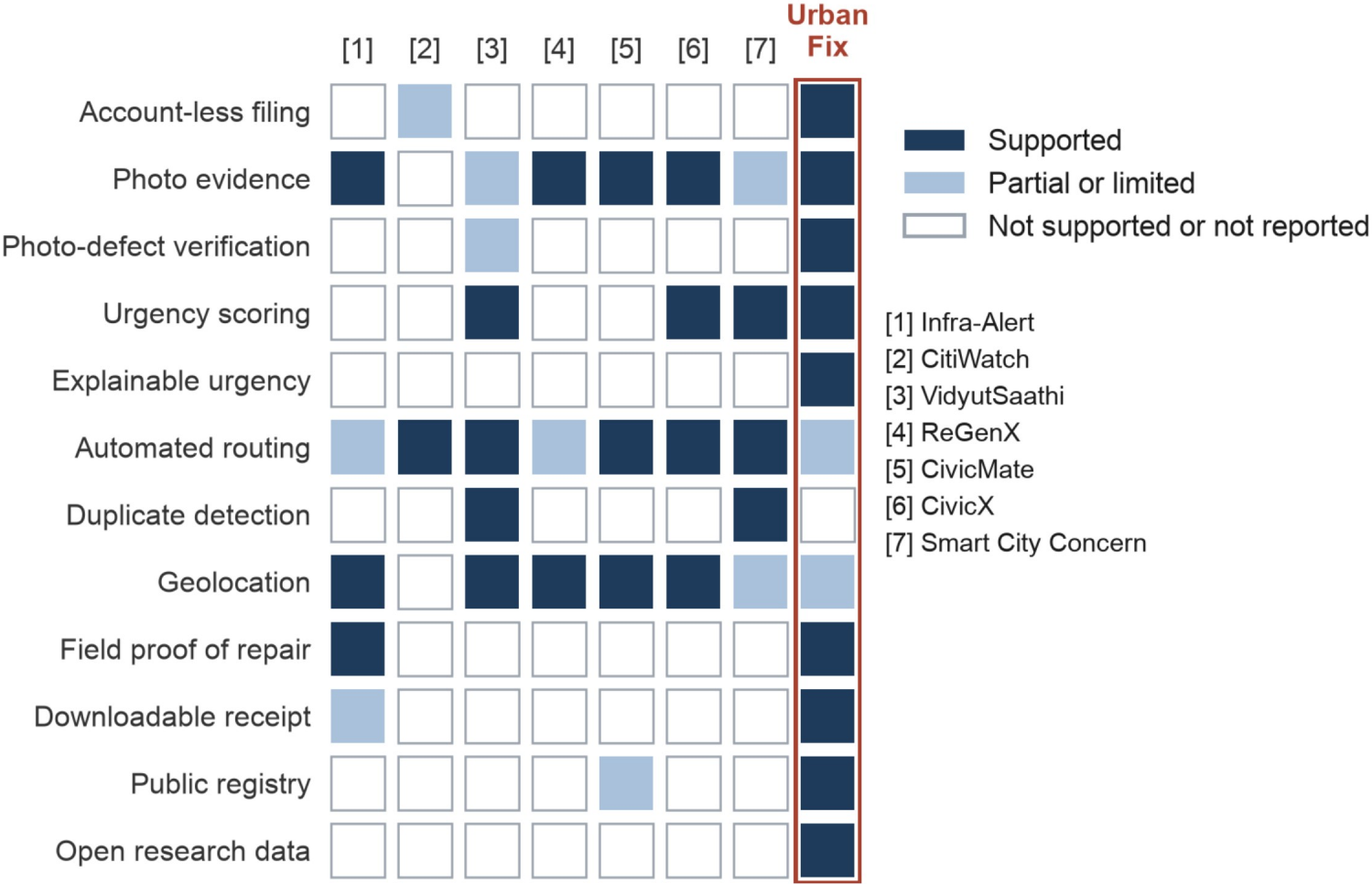


Source: DARPG reply in the Rajya Sabha on the CPGRAMS portal (PIB, 5 Feb 2026). Municipal and civic complaints fall under State/UT governments.

02

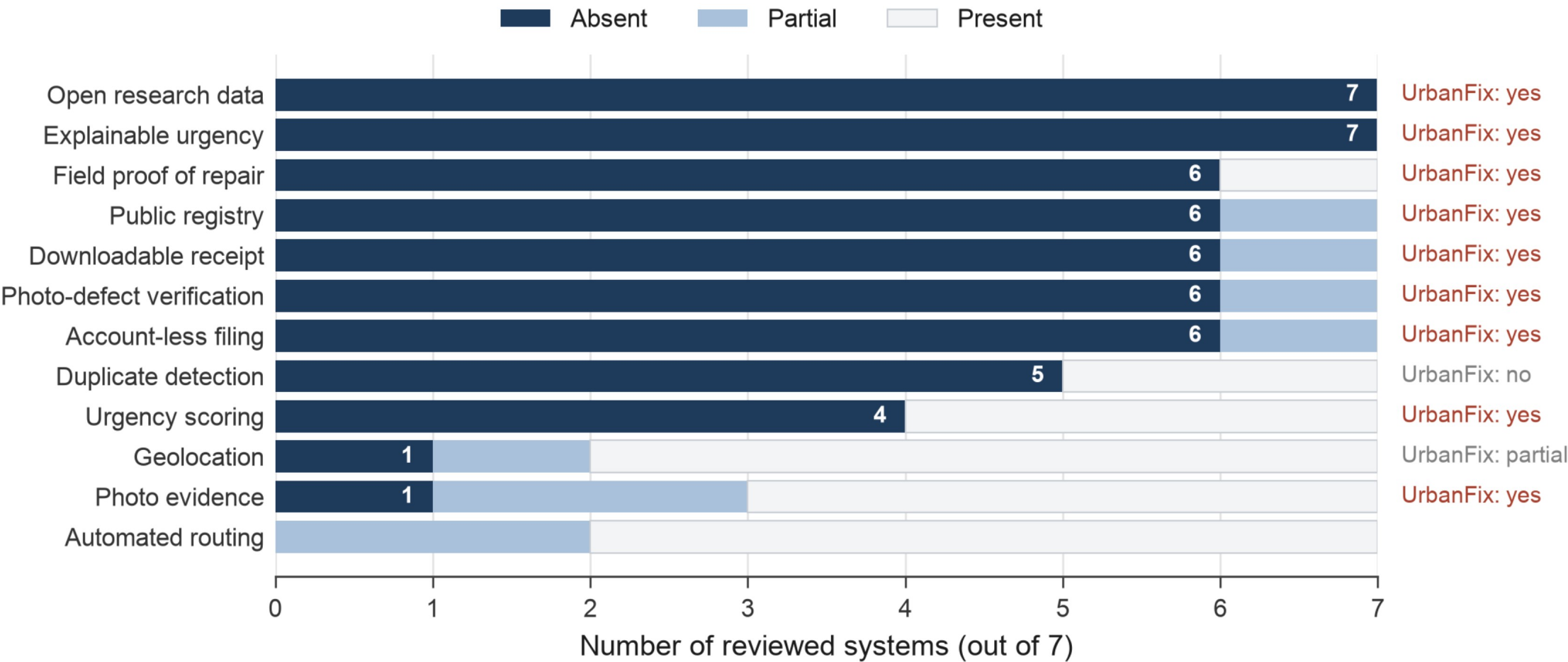
Gap

PRIOR SYSTEMS



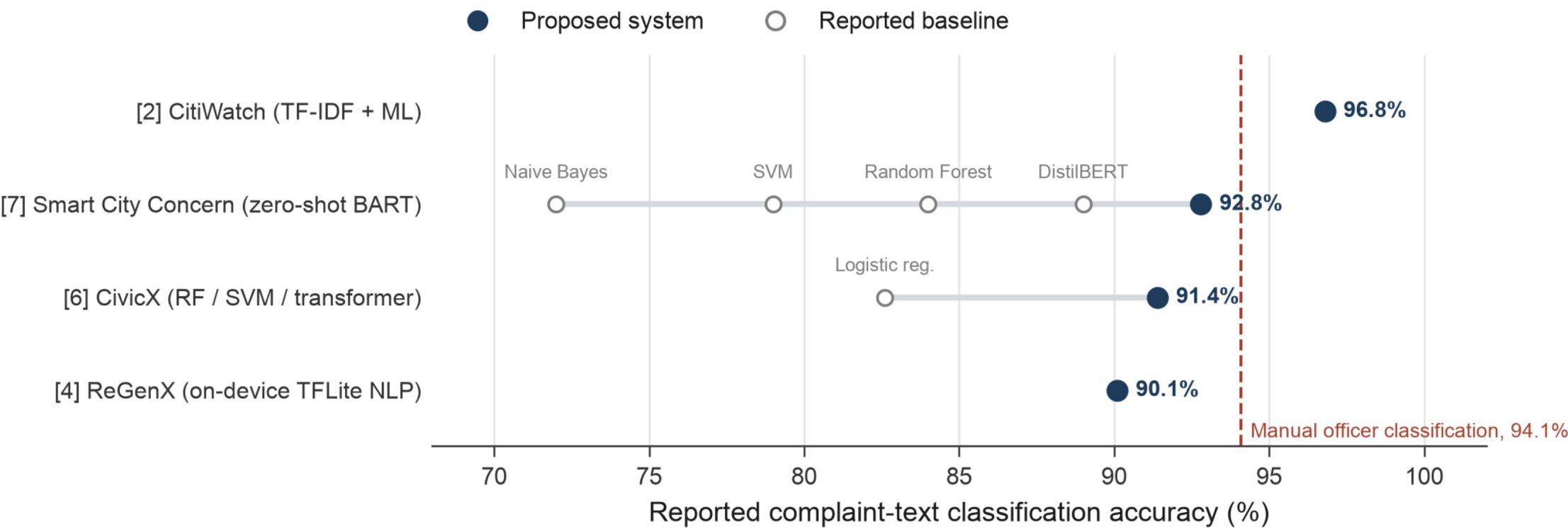
Source: authors' literature survey of [1] to [7]; ratings follow each paper's reported implementation.

CAPABILITY GAPS



Derived from the feature coverage matrix (Fig. 2.1).

TEXT-ONLY AI IN PRIOR WORK



Source: results reported in [2], [4], [6] and [7]; manual baseline from [6].

03

System

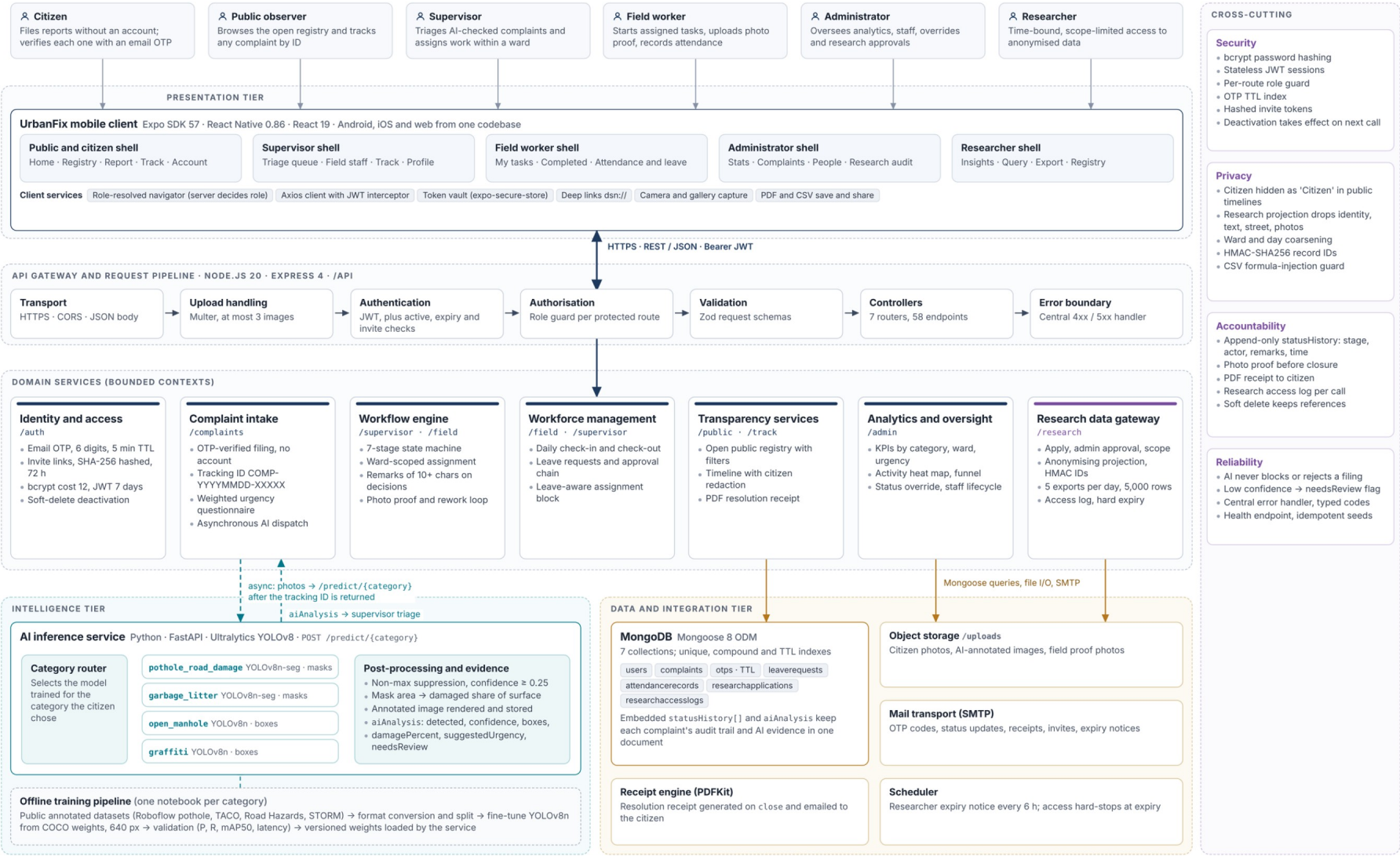
GOVERNANCE LOOP



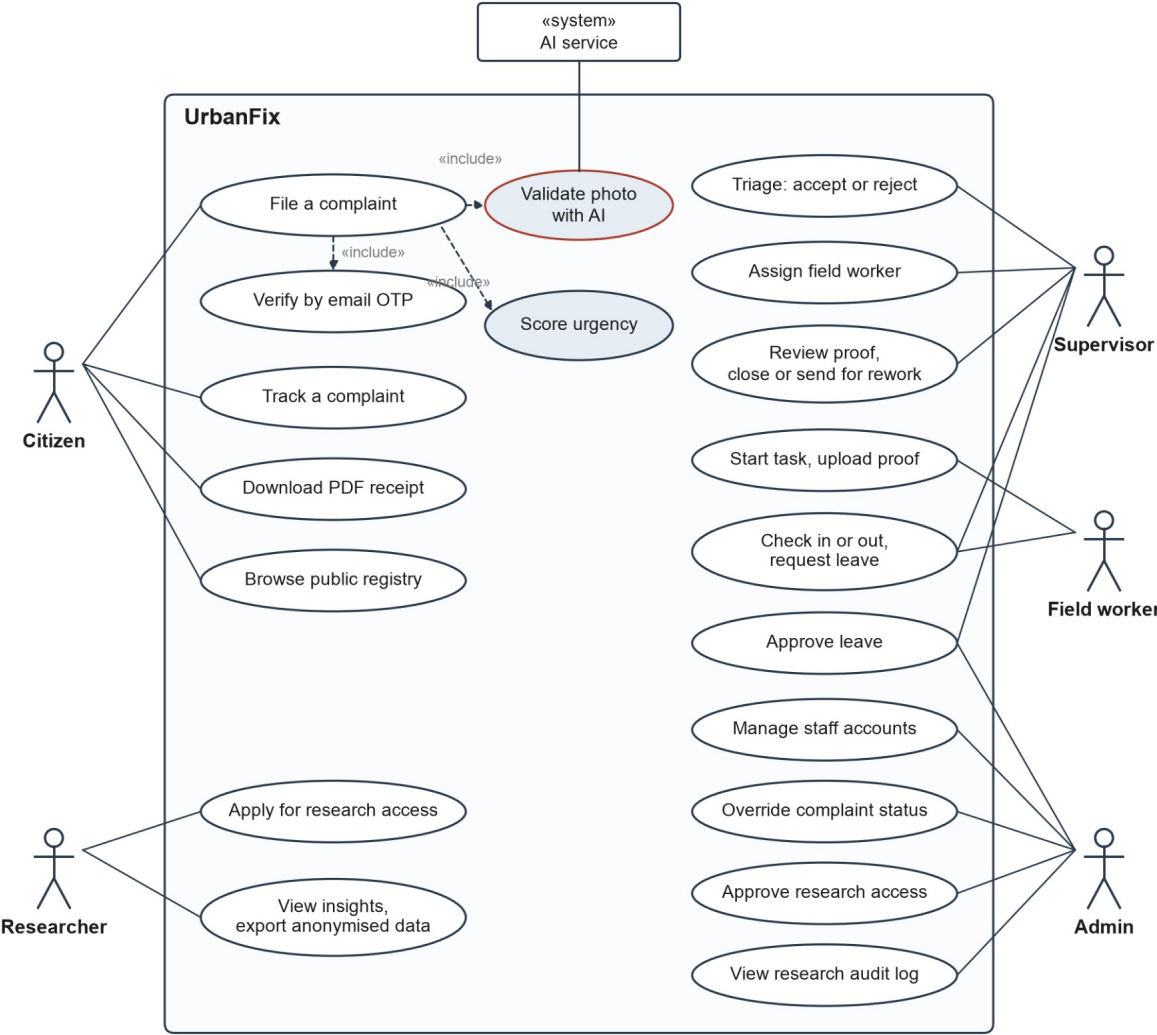
ARCHITECTURE

UrbanFix: system architecture

Layered container view. Solid arrows: synchronous requests. Dashed arrows: asynchronous or offline flows.



USE CASES



ACCESS CONTROL

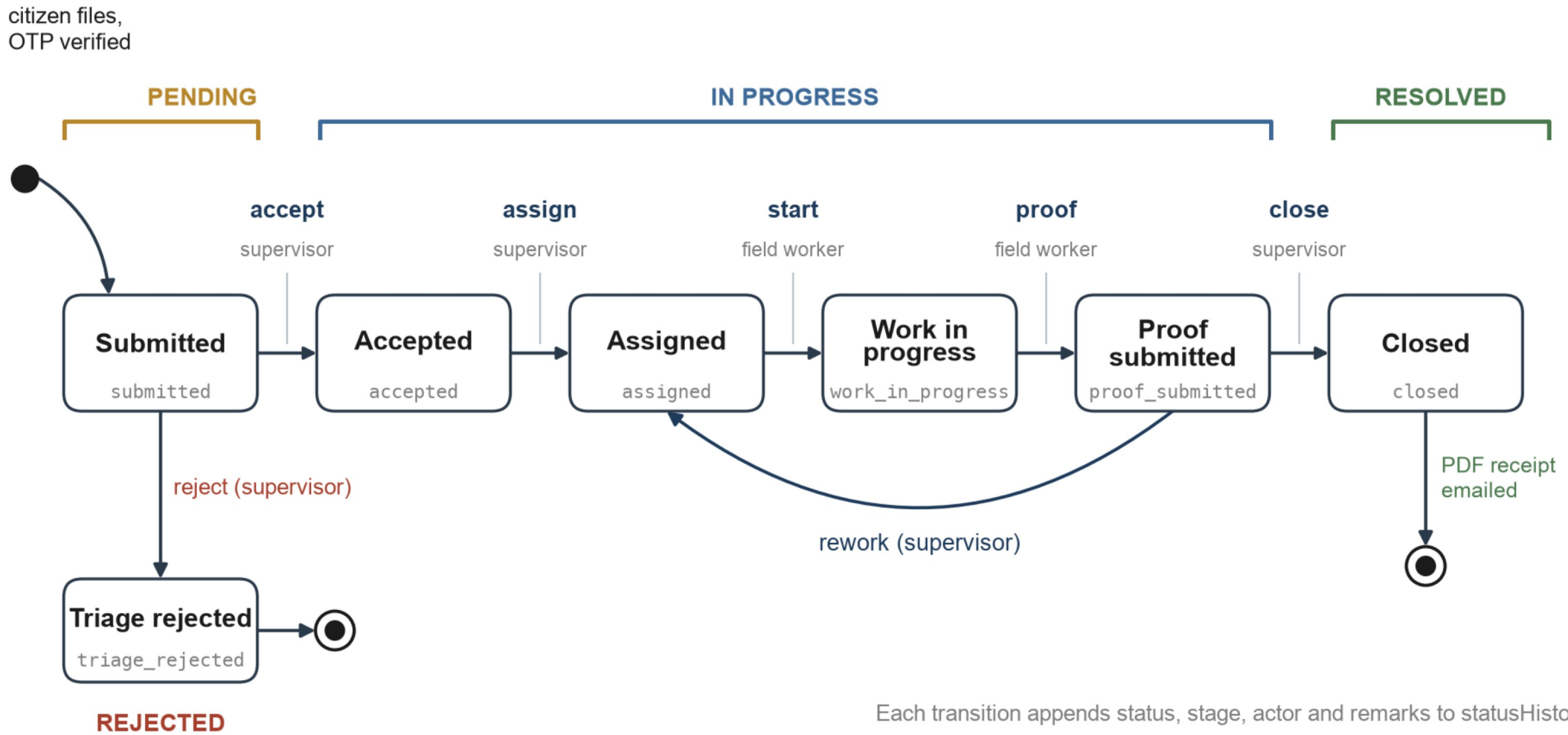
Role-based access control matrix

Enforced on the server by a JWT check and a role guard on every protected route; the client only decides which screens to draw.

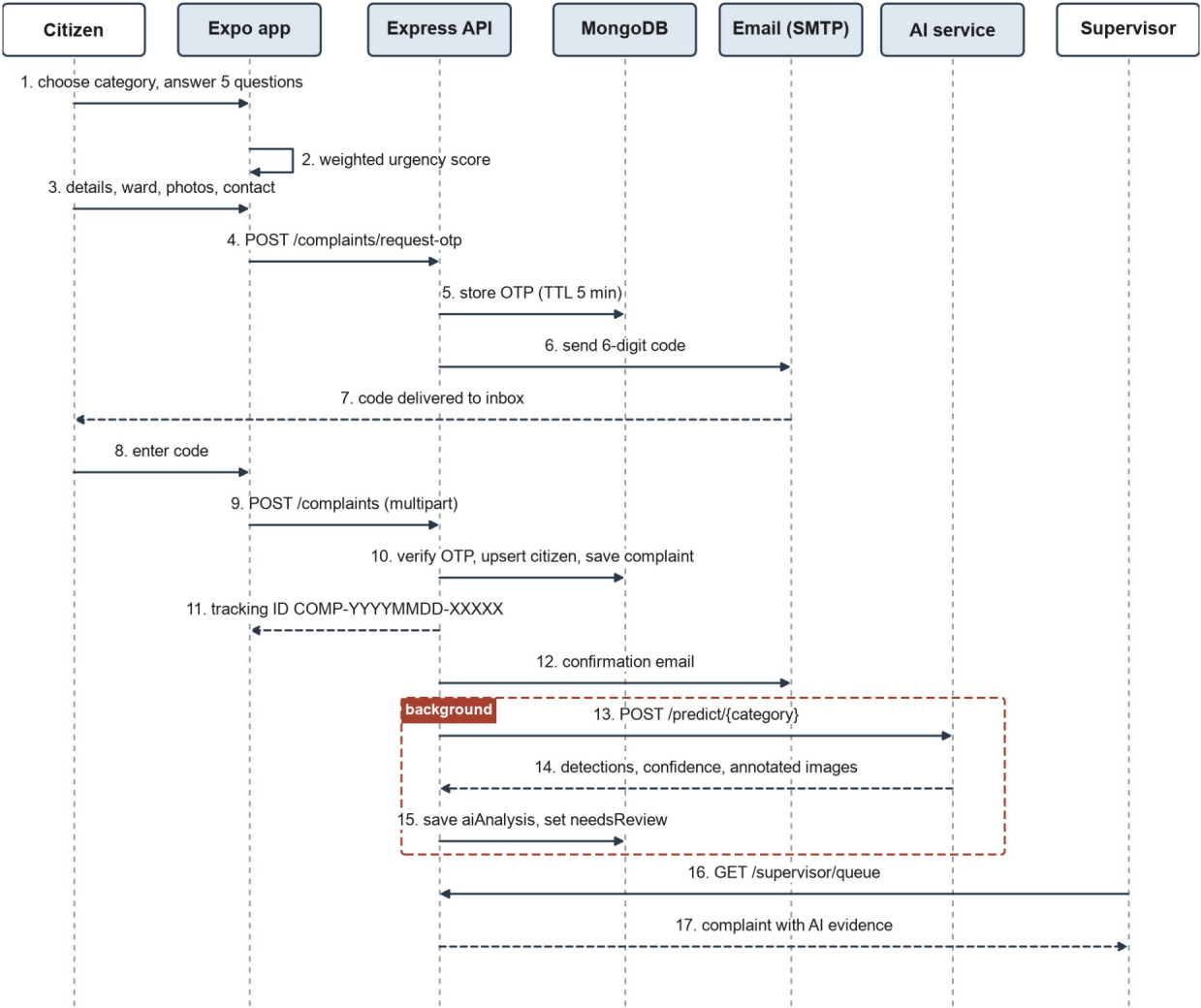
Capability	Public no account	Citizen registered	Field worker	Supervisor	Admin	Researcher
TRANSPARENCY						
Browse public registry and city counters	●	●	●	●	●	●
Track any complaint, download receipt	●	●	●	●	●	●
REPORTING						
File a complaint with email OTP	●	●	×	×	×	×
View own complaint history	×	●	×	×	×	×
WORKFLOW						
Accept or reject at triage	×	×	×	●	●	×
Assign a field worker	×	×	×	ⓘ own ward	● any ward	×
Start task, submit photo proof	×	×	ⓘ assignee only	×	×	×
Close with receipt, or send for rework	×	×	×	●	●	×
Override public status	×	×	×	×	●	×
WORKFORCE						
Check in and out, request leave	×	×	●	●	×	×
Decide leave requests	×	×	×	ⓘ field staff	ⓘ supervisors	×
Invite, edit, deactivate staff	×	×	×	×	●	×
ANALYTICS AND RESEARCH						
Operational statistics	×	×	ⓘ own tasks	ⓘ queue stats	● city-wide	×
Apply for research access	●	×	×	×	×	×
Approve applications, read research audit log	×	×	×	×	●	×
Research insights and grouped queries	×	×	×	×	●	● until expiry
Row-level anonymised records and export	×	×	×	×	●	ⓘ if scope allows

● allowed ⓘ allowed with a scope restriction × denied (HTTP 401 or 403)

COMPLAINT LIFECYCLE



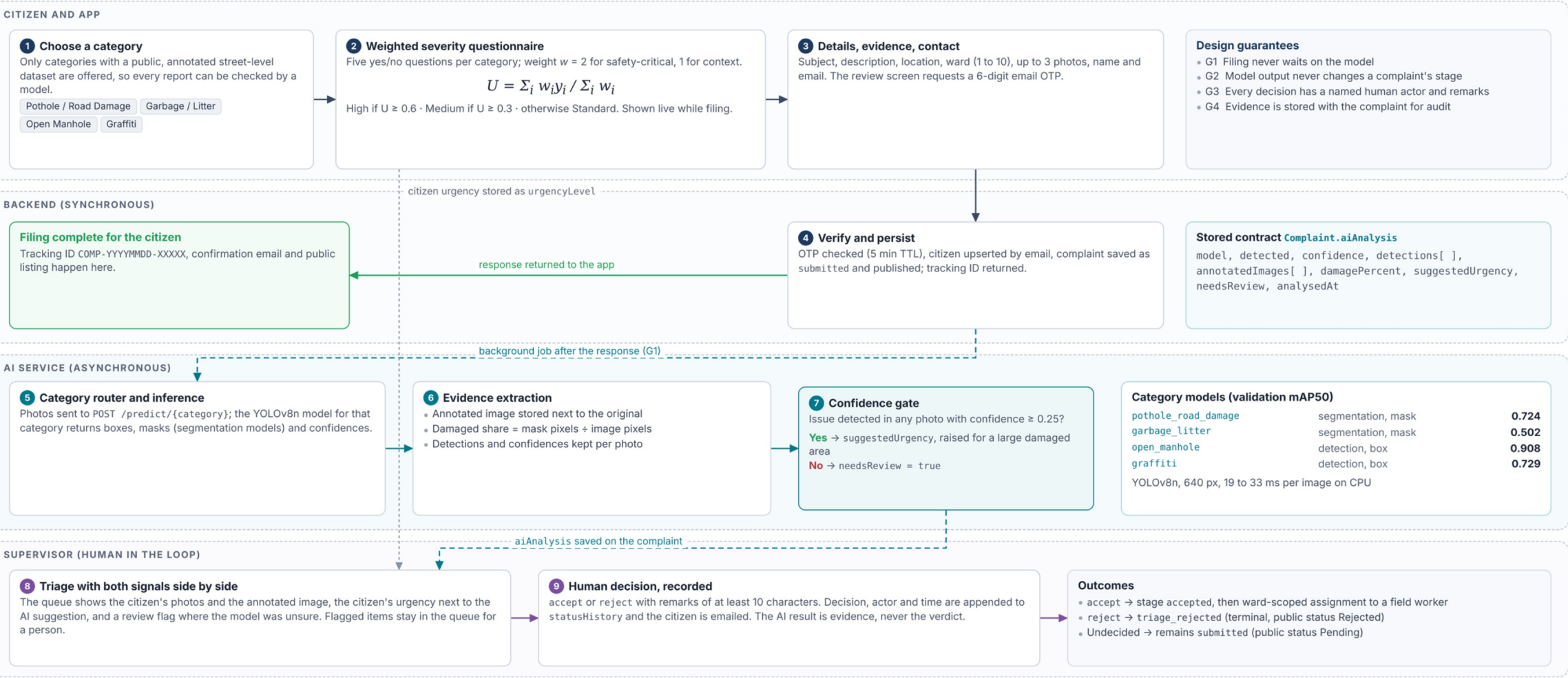
SUBMISSION FLOW



AI VALIDATION PIPELINE

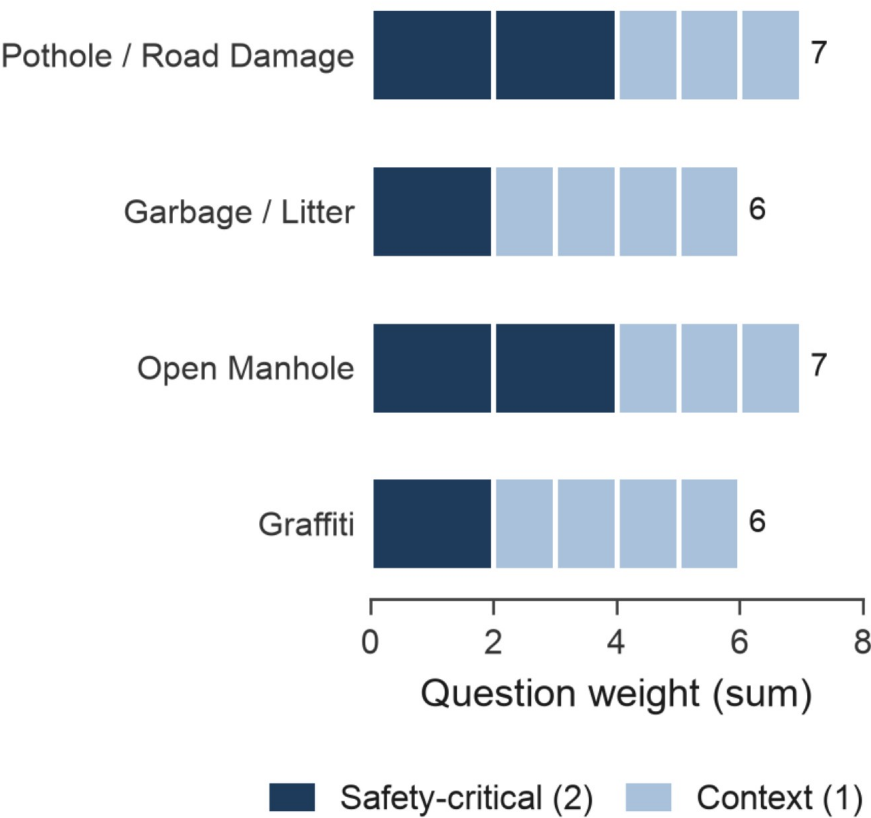
AI validation and prioritisation pipeline

Two independent urgency signals reach a human decision-maker. The filing completes before inference starts, and the model can flag a report but never reject it.

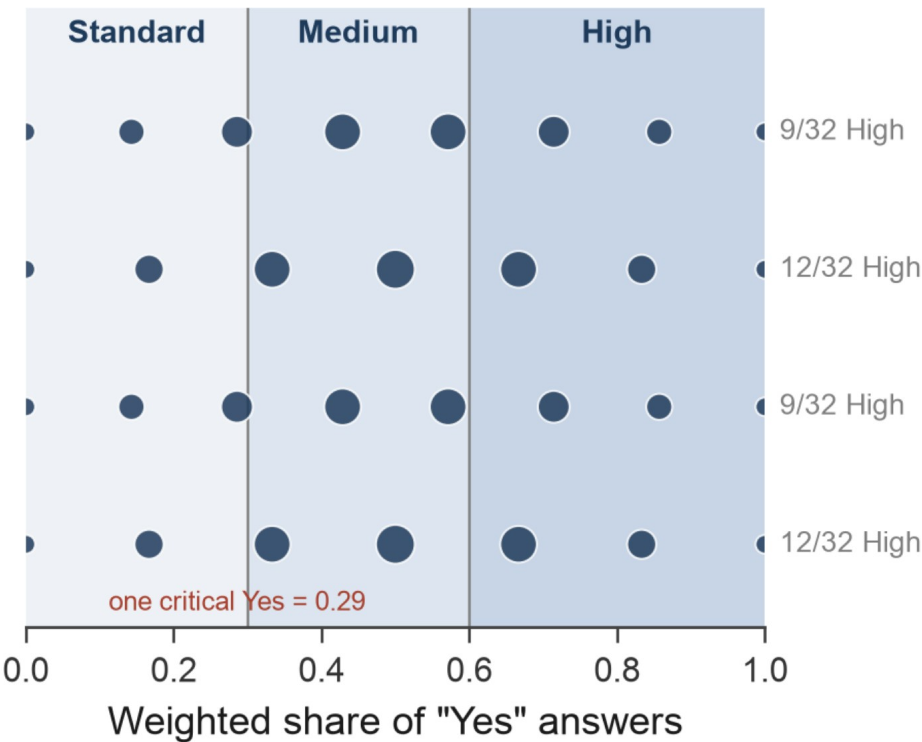


URGENCY SCORING

(a) Weights of the five questions



(b) Scores reachable from all 32 answer sets



Marker area is proportional to the number of answer sets giving that score. Thresholds: High at 0.6 or above, Medium at 0.3 or above (mobile/src/utils/urgency.js).

DATA DISCLOSURE

Tiered data disclosure model

What each audience receives from one complaint record. Every view is produced server-side by a fixed projection; clients never receive the full document.

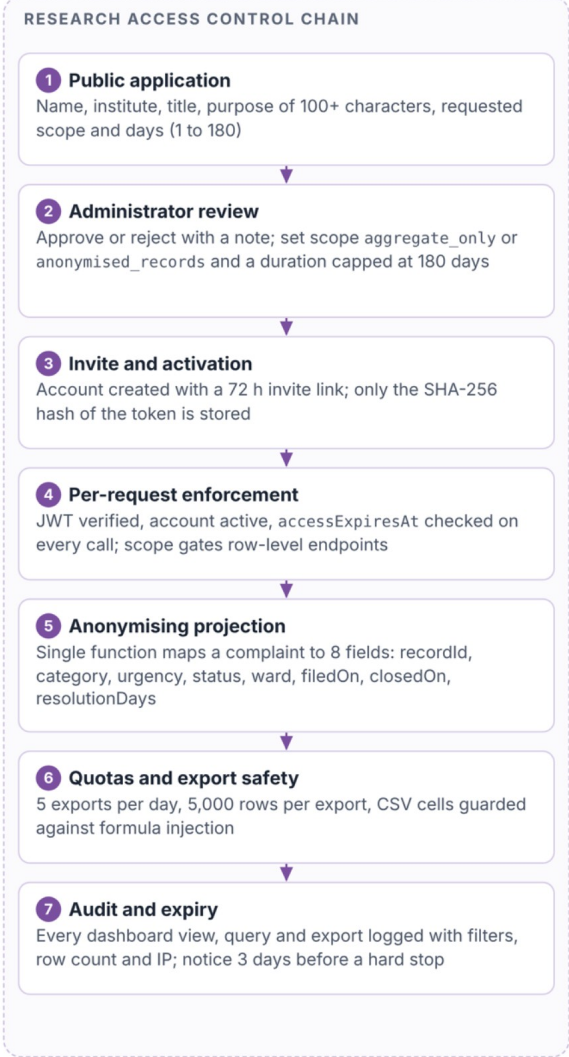
Attribute of a complaint	Administrator	Supervisor and field worker	Public registry no login	Public tracker by tracking ID	Researcher anonymised records	Researcher aggregate only
IDENTITY						
Citizen name, email, phone	●	×	×	×	×	×
				shown as "Citizen"		
Assigned and assigning staff	●	●	×	●	×	×
				names in timeline		
Record identifier	● tracking ID	● tracking ID	● tracking ID	● tracking ID	● HMAC-SHA256, 12 hex	×
CONTENT						
Title and free-text description	●	●	●	●	×	×
Photos: citizen, AI-annotated, proof	●	●	●	●	×	×
Street-level location	●	●	●	●	● coarsened to ward	×
GOVERNANCE ATTRIBUTES						
Category, urgency, status, ward	●	●	●	●	●	● grouped counts
Filing and closing time	●	●	●	●	● day precision	● monthly series
Resolution time	●	●	●	●	● derived, days	● averages
Status history with remarks	●	●	● without actors	● citizen redacted	×	×

● full value ● transformed (coarsened, pseudonymised or aggregated) × withheld

Least privilege by audience
Staff see what the work needs; triage hides the filer's identity because the issue, not the person, is being judged.

Public by default, personal never
Every complaint is published on filing with its full timeline; identity fields are removed by the query itself, not by the client.

Research data is a product, not a dump
Researchers receive a governed, logged, expiring projection that cannot be joined back to a citizen without the server secret.



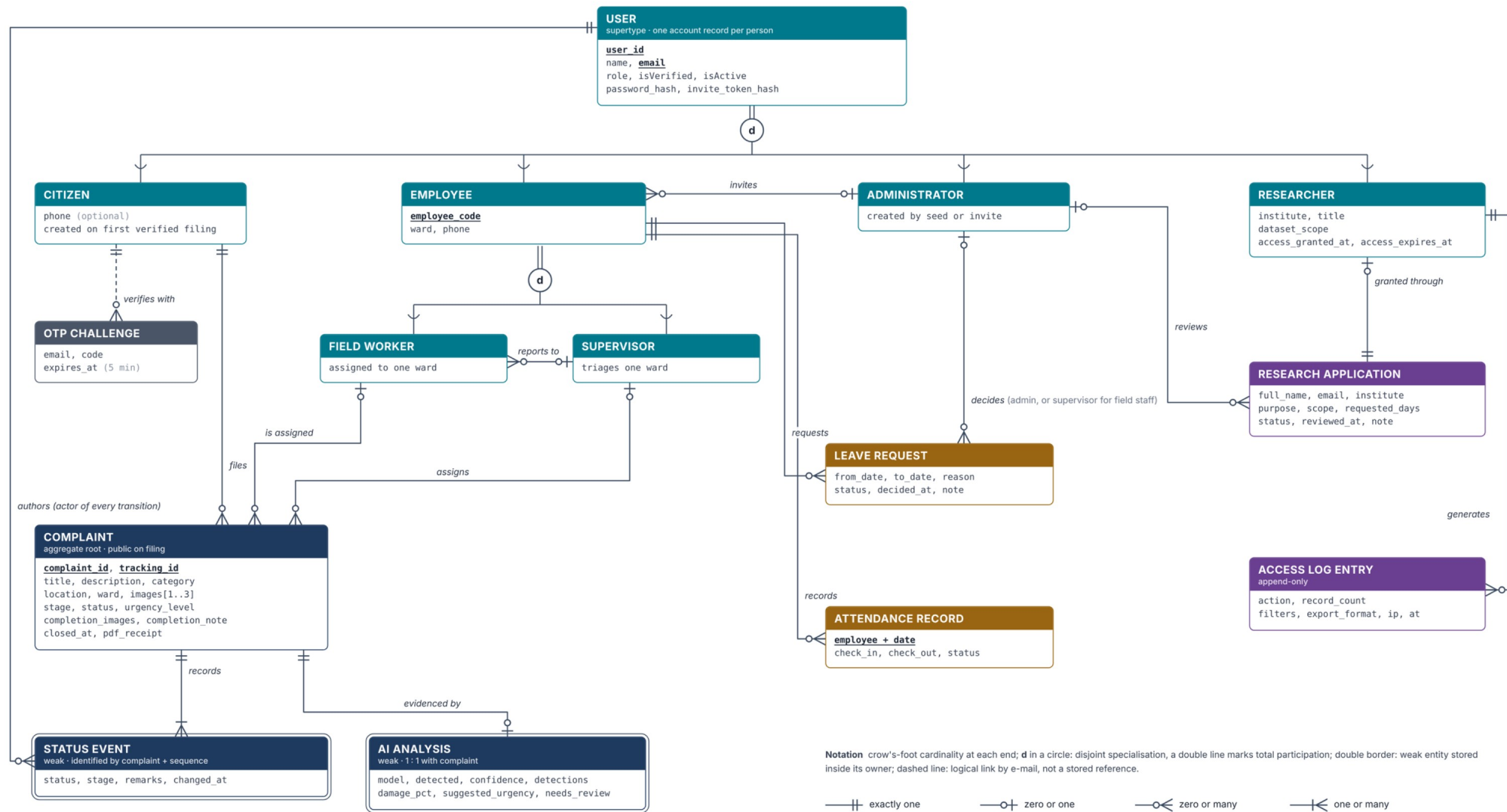
04

Data model

CONCEPTUAL MODEL

UrbanFix conceptual data model (EER)

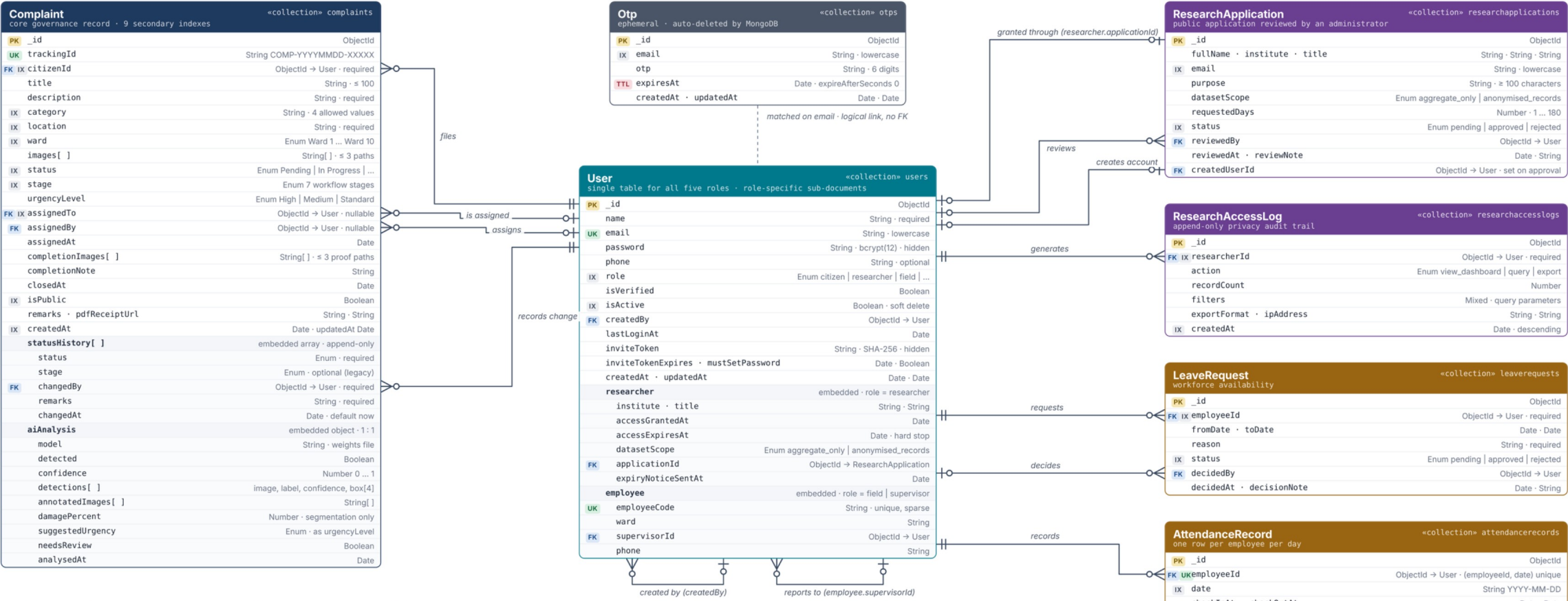
Role specialisation of the single User entity, the complaint aggregate with its weak entities, and the workforce and research sub-models. Underlined attributes identify an entity.



PHYSICAL MODEL

UrbanFix physical data model

MongoDB 8 collections as defined by the Mongoose schemas in backend/models. Shaded rows are embedded documents; every createdAt / updatedAt pair is maintained by the ODM.



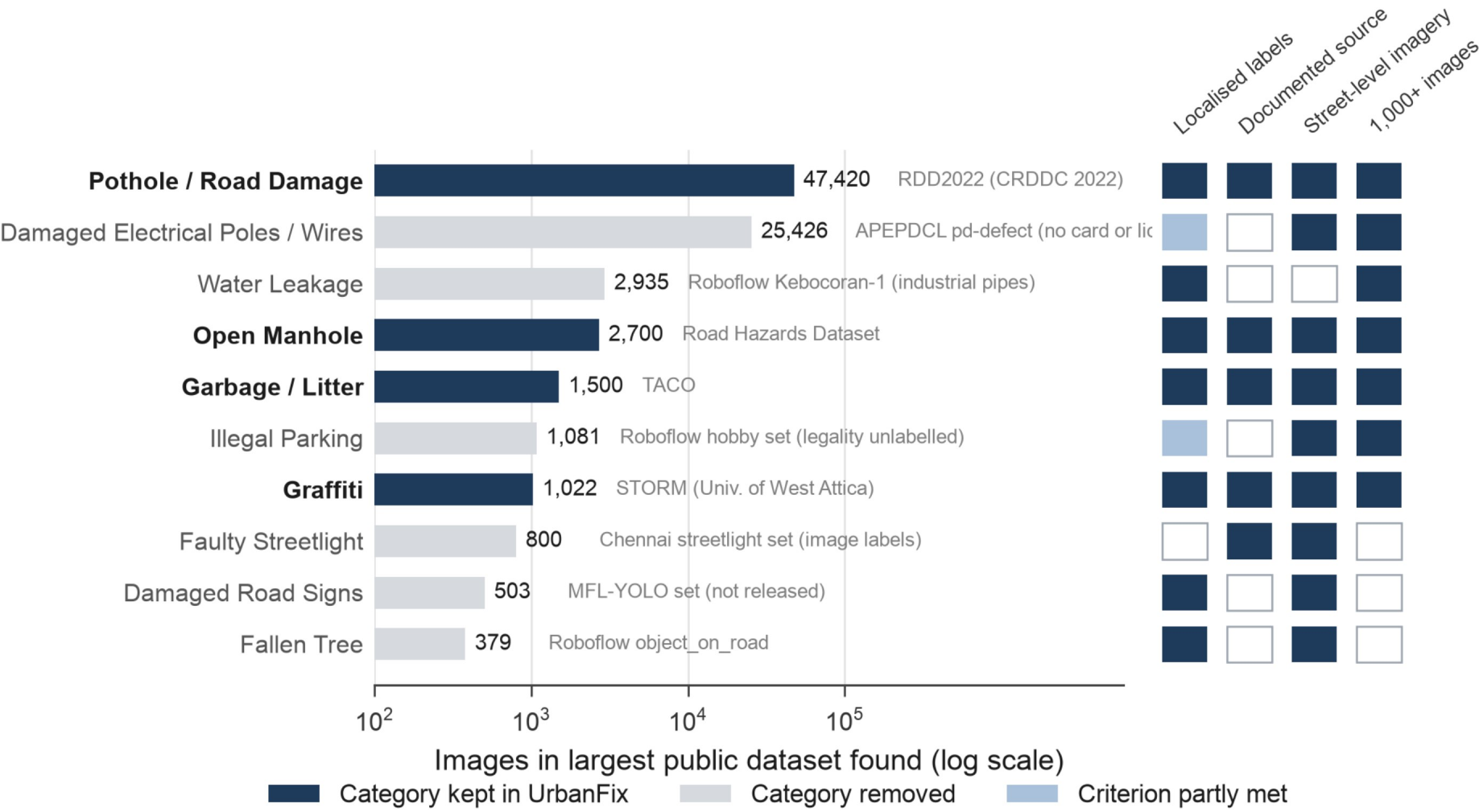
Keys PK primary FK reference (Objectid, Mongoose ref) UK unique IX secondary index TTL expiry index

—|| exactly one —|+ zero or one —|< zero or many
Crow's-foot notation. The marker at each end gives how many records on that side take part in one instance of the relationship.

05

AI validation

DATASETS



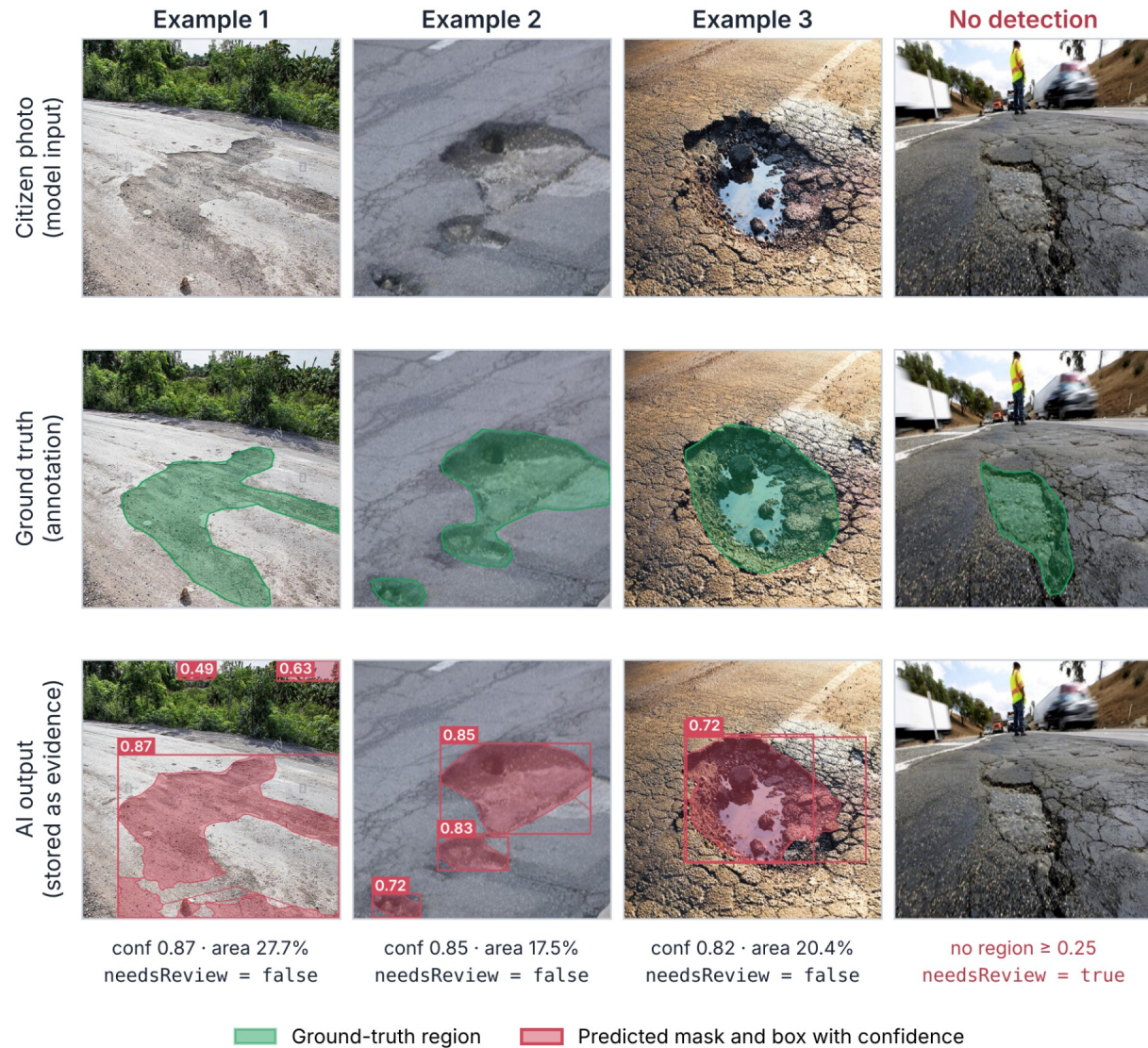
Sources: RDD2022, TACO, Road Hazards Dataset, STORM, Hugging Face, Roboflow Universe and arXiv:2407.01117. Sizes are image counts as published.

AI OUTPUT

Pothole

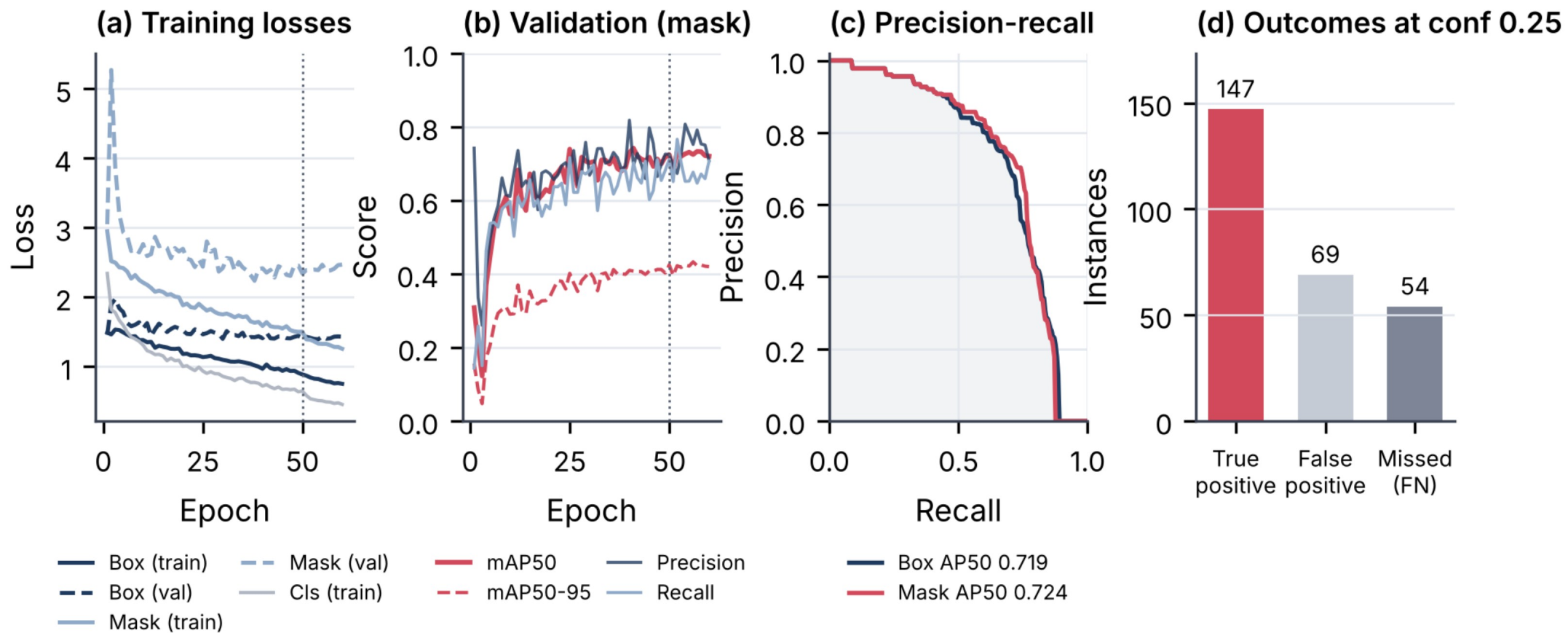
Segmentation · mask mAP50 0.724

Pothole / Road Damage · Instance segmentation (YOLOv8n)



POTHOLE · EVALUATION

Pothole / Road Damage · Instance segmentation · 60 epochs, best 50, 41 min training · 33 ms CPU / 11 ms MPS per image



AI OUTPUT

Litter

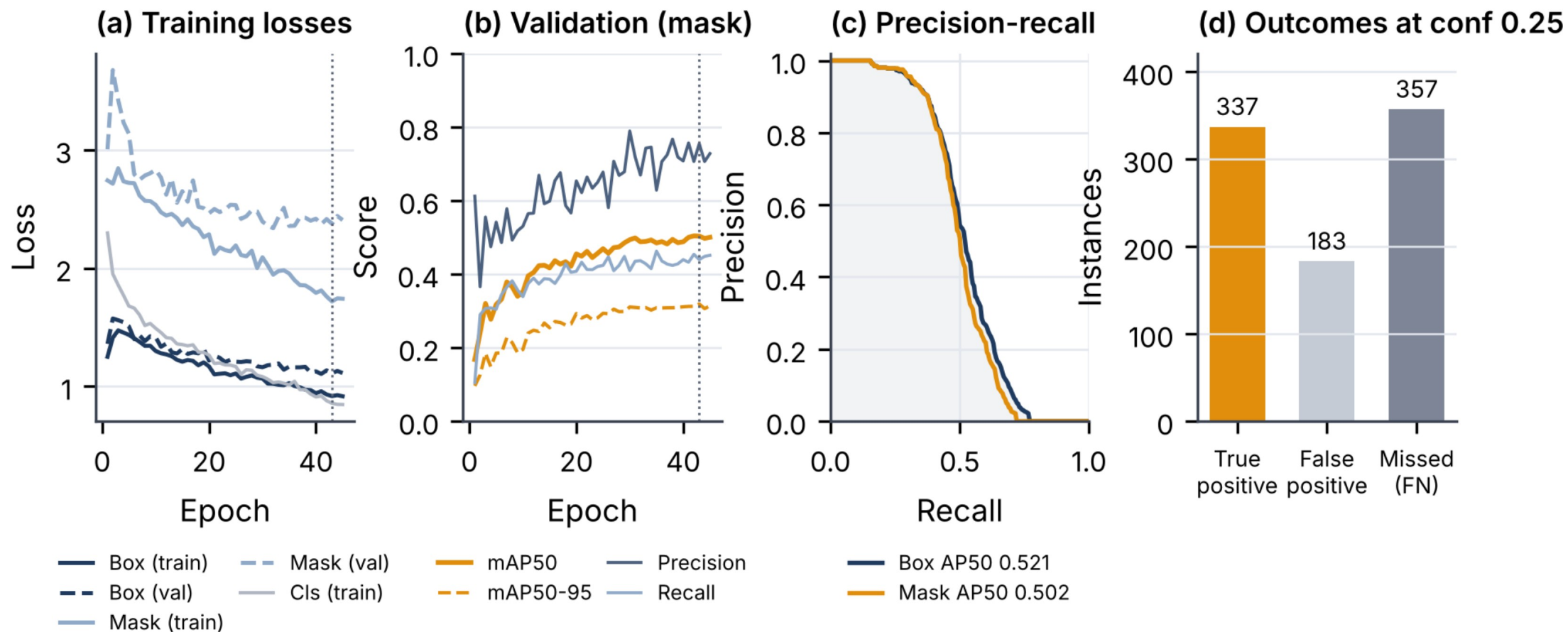
Segmentation · mask mAP50 0.502

Garbage / Litter · Instance segmentation (YOLOv8n)



LITTER · EVALUATION

Garbage / Litter · Instance segmentation · 45 epochs, best 43, 52 min training · 24 ms CPU / 20 ms MPS per image



AI OUTPUT

Open manhole

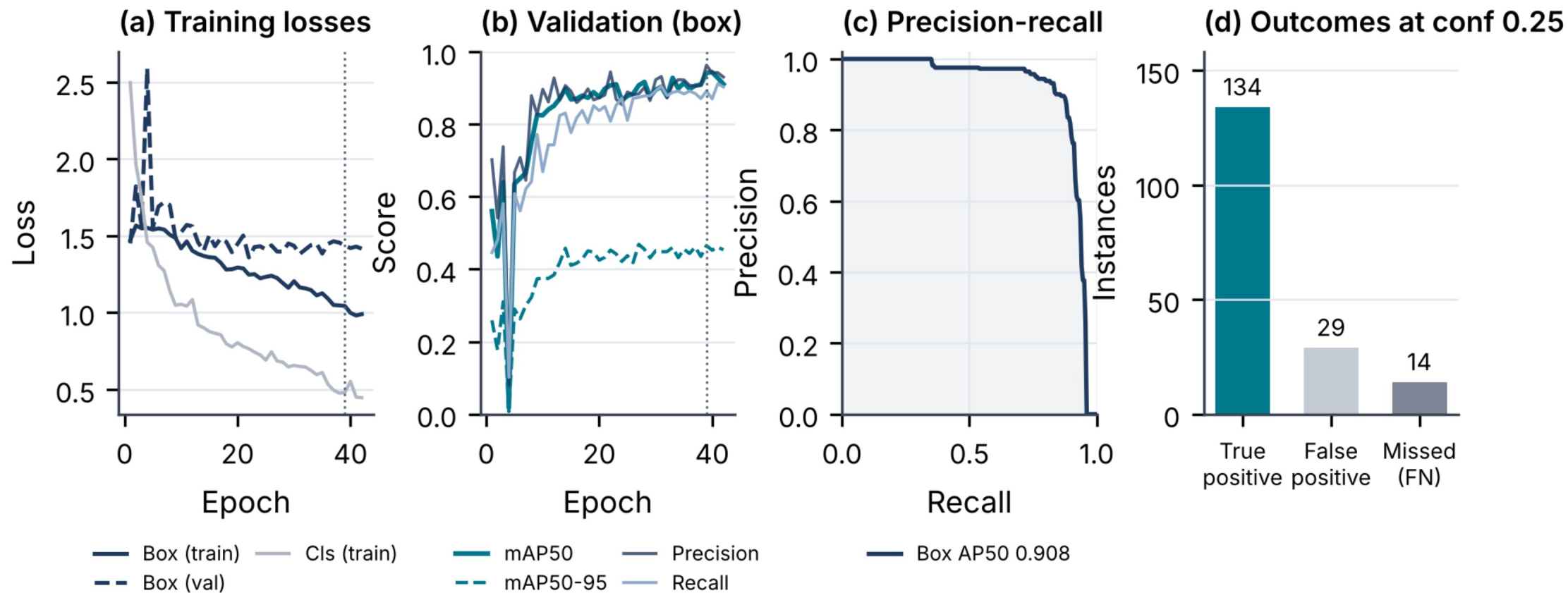
Detection · box mAP50 0.908

Open Manhole · Object detection (YOLOv8n)

	Example 1	Example 2	Example 3	No detection
Citizen photo (model input)				
Ground truth (annotation)				
AI output (stored as evidence)				
	conf 0.84 · 1 region needsReview = false	conf 0.82 · 1 region needsReview = false	conf 0.80 · 1 region needsReview = false	no region ≥ 0.25 needsReview = true
	 Ground-truth region			 Predicted box with confidence

OPEN MANHOLE · EVALUATION

Open Manhole · Object detection · 42 epochs, best 39, 34 min training · 19 ms CPU / 8 ms MPS per image



AI OUTPUT

Graffiti

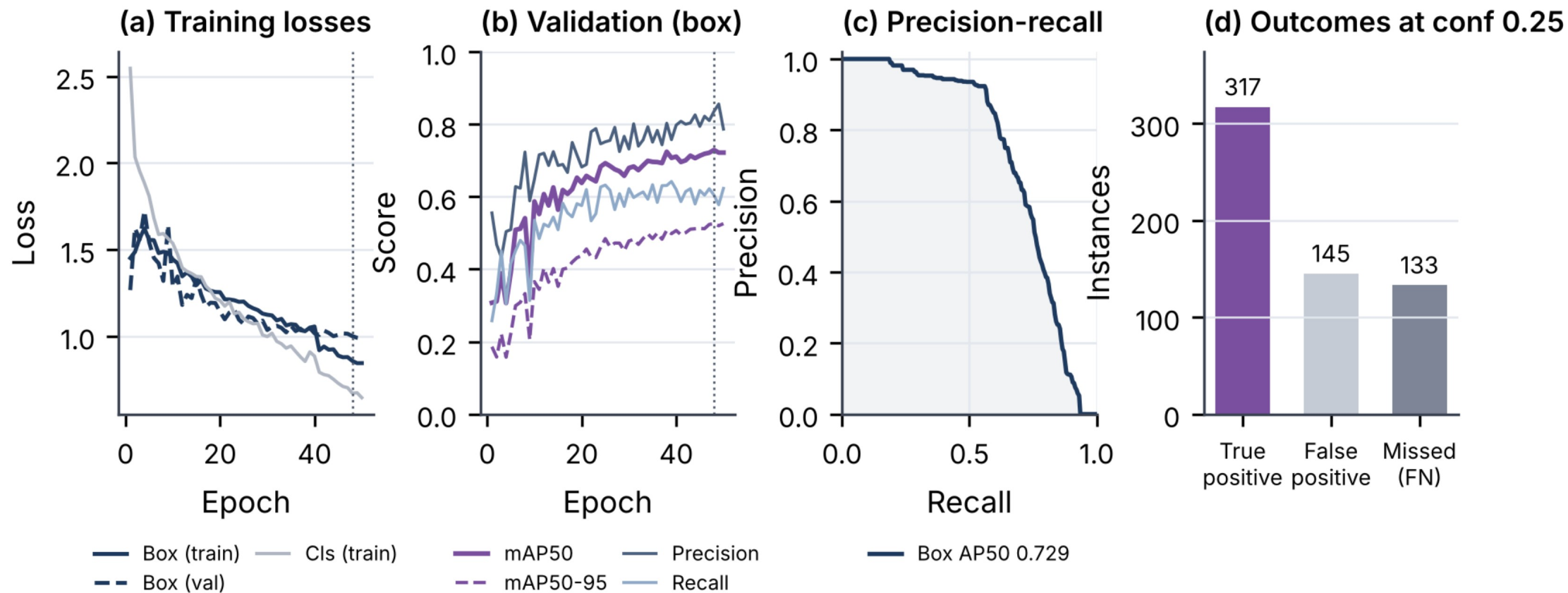
Detection · box mAP50 0.729

Graffiti · Object detection (YOLOv8n)

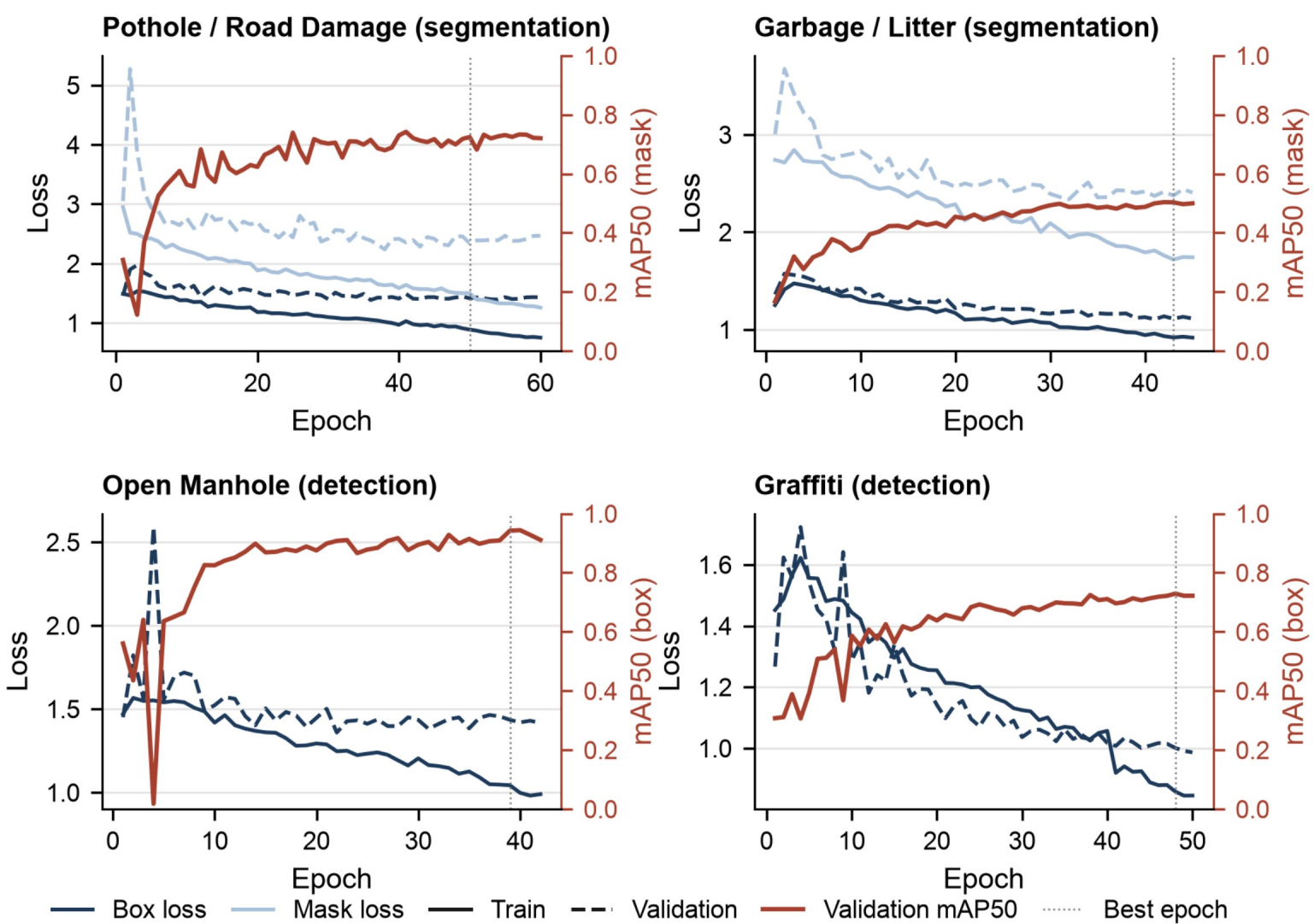
	Example 1	Example 2	Example 3	No detection
Citizen photo (model input)				
Ground truth (annotation)				
AI output (stored as evidence)				
	conf 0.91 · 5 regions needsReview = false	conf 0.74 · 1 region needsReview = false	conf 0.94 · 1 region needsReview = false	no region ≥ 0.25 needsReview = true
	 Ground-truth region			 Predicted box with confidence

GRAFFITI · EVALUATION

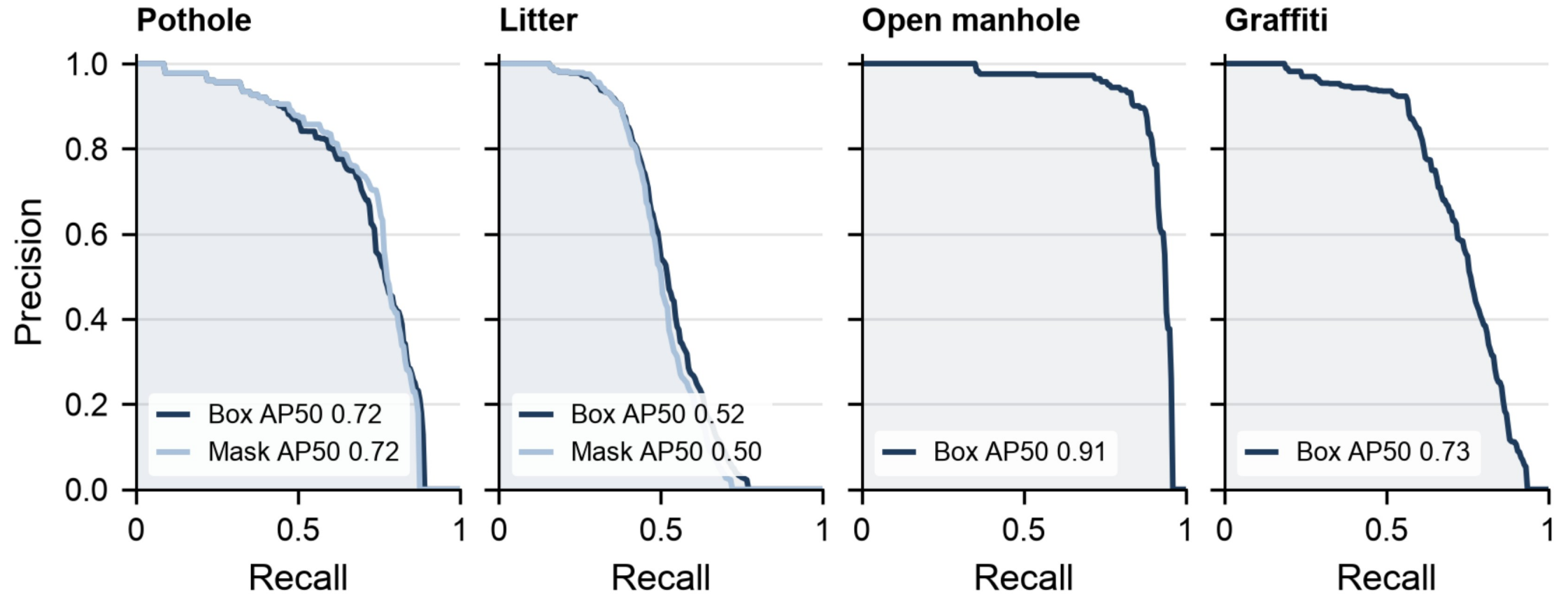
Graffiti · Object detection · 50 epochs, best 48, 29 min training · 20 ms CPU / 6 ms MPS per image



TRAINING · ALL MODELS



PRECISION-RECALL



CONFUSION MATRICES

Pothole / Road Damage
P 0.68, R 0.73

Predicted	background	defect
	background	defect
	TP 147 0.73	FP 69
	FN 54 0.27	n/a
	defect	background
	True	

Garbage / Litter
P 0.65, R 0.49

Predicted	background	defect
	background	defect
	TP 337 0.49	FP 183
	FN 357 0.51	n/a
	defect	background
	True	

Open Manhole
P 0.82, R 0.91

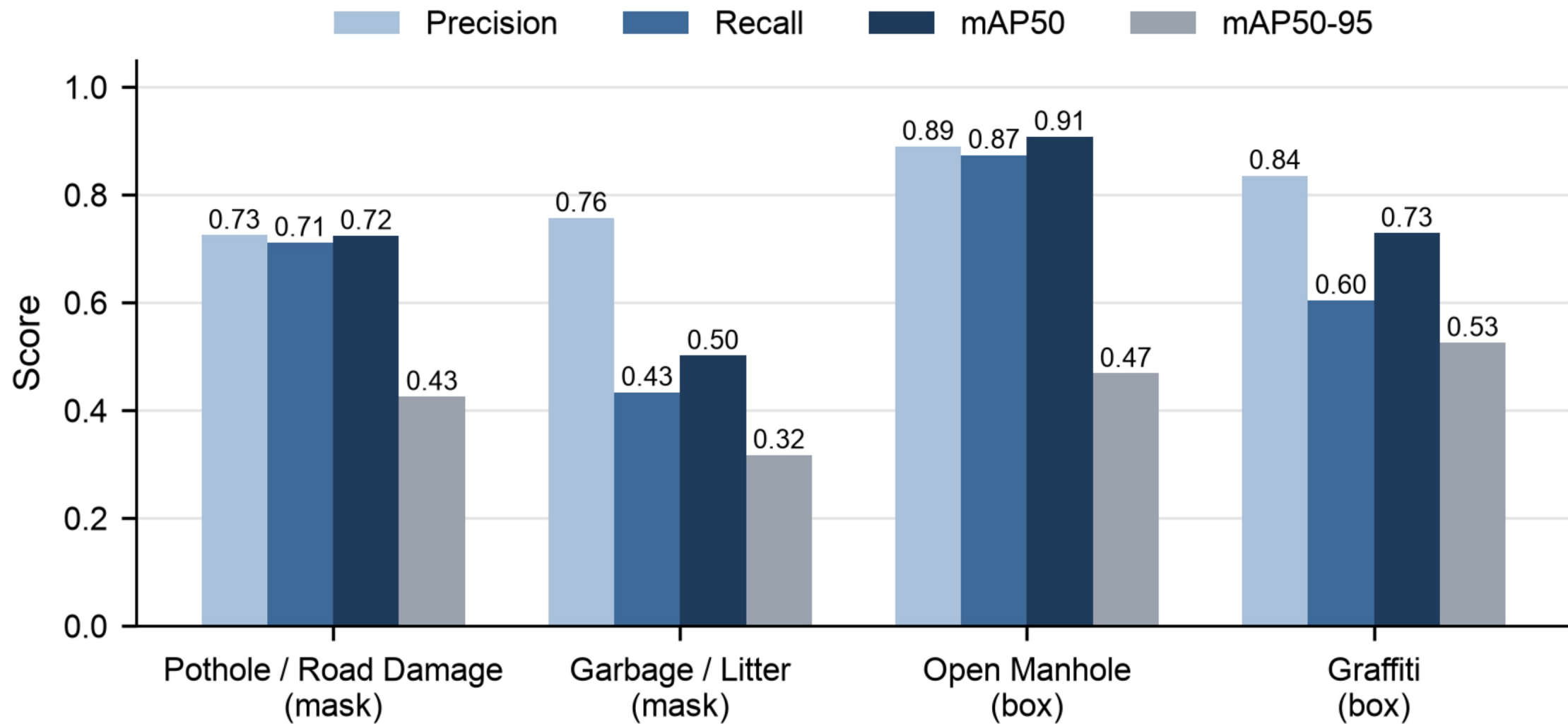
Predicted	background	defect
	background	defect
	TP 134 0.91	FP 29
	FN 14 0.09	n/a
	defect	background
	True	

Graffiti
P 0.69, R 0.70

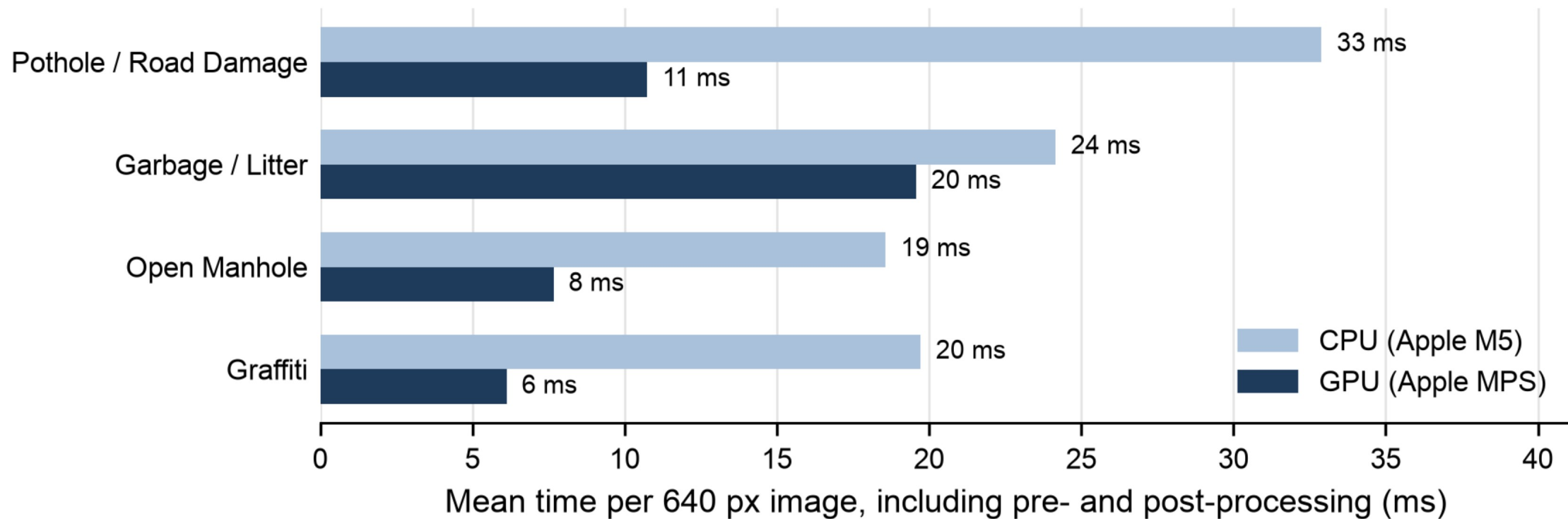
Predicted	background	defect
	background	defect
	TP 317 0.70	FP 145
	FN 133 0.30	n/a
	defect	background
	True	

Instance counts at confidence 0.25, IoU 0.45. The defect column is normalised by true defects. Detectors have no background-background cell.

MODEL COMPARISON



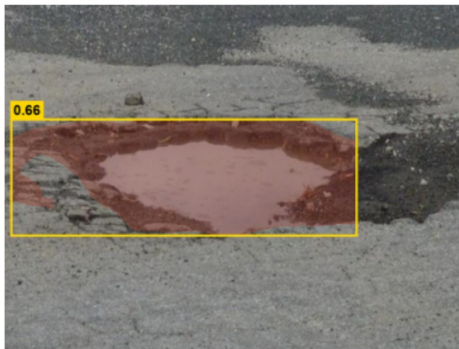
LATENCY



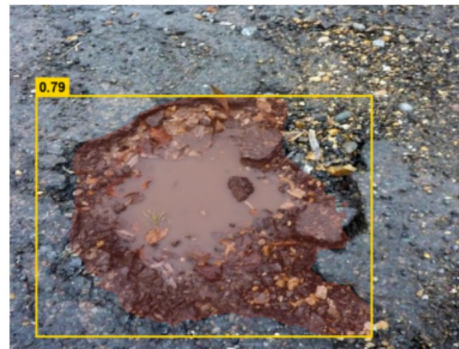
Mean over 40 validation images after 3 warm-up runs, batch size 1.

DETECTIONS

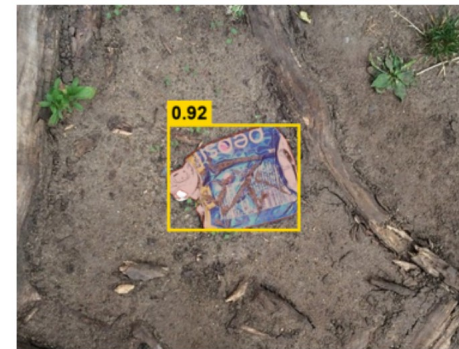
Pothole



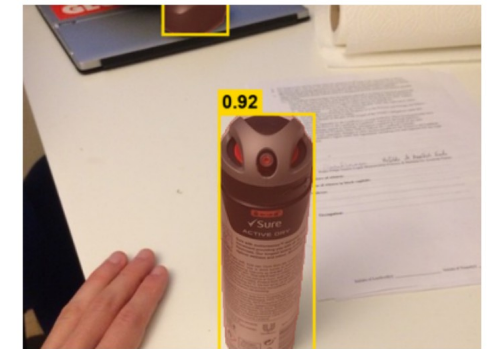
Pothole



Litter



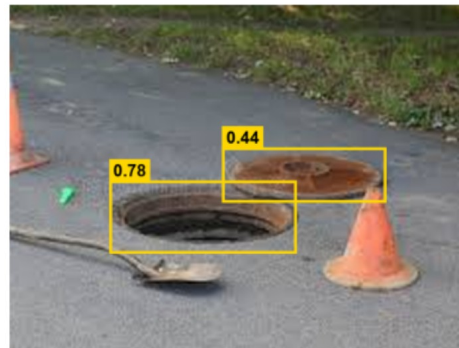
Litter



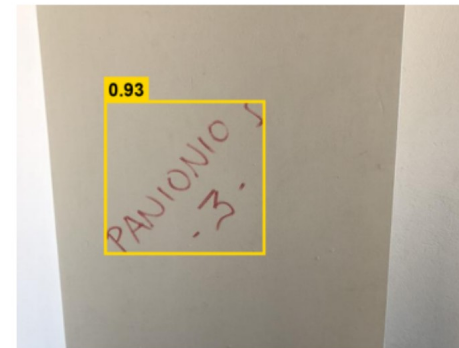
Open manhole



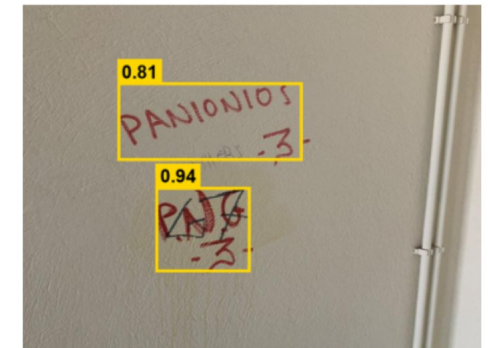
Open manhole



Graffiti



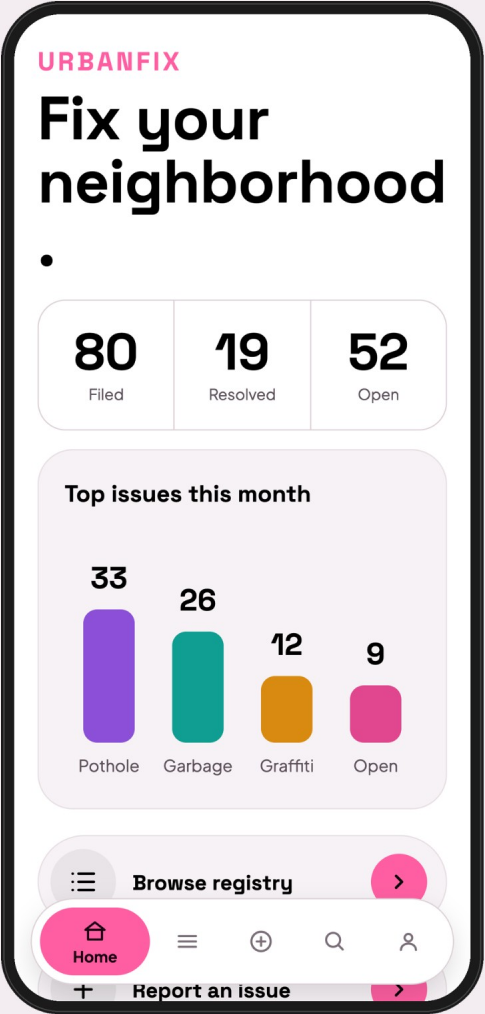
Graffiti



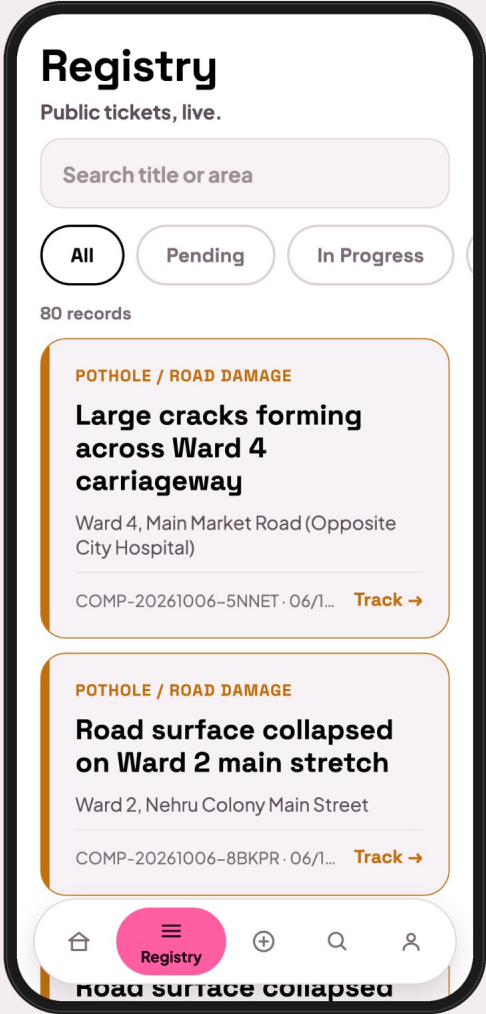
Yellow boxes: detections with confidence. Red fill: predicted masks (segmentation models). Images from each validation set.

06

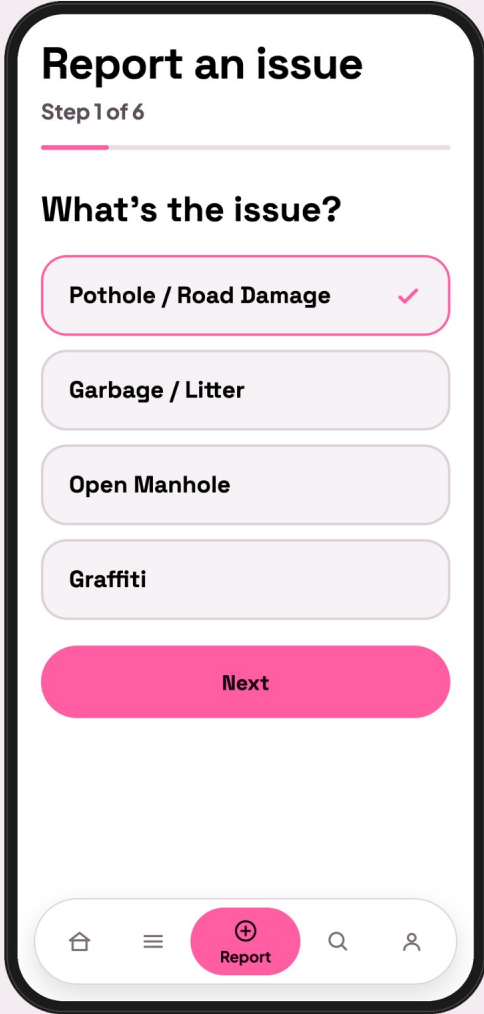
The app



Home



Registry



Report

<

Report an issue

Question 1 of 5

POTHOLE / ROAD DAMAGE

Has this pothole/damage caused an accident or vehicle damage?

< Swipe to answer >

✕

✓

• Priority

Standard

🏠

☰

⊕

Report

🔍

👤

Questionnaire

Track

Enter your tracking ID.

COMP-20251020-1DJ7N

🔍

Find my complaint

Cluster of potholes slowing traffic on Ward 3

• Resolved

COMP-20251020-1DJ7N

QUESTIONNAIRE

Has this pothole/damage caused an accident or vehicle damage?

NO

Is the pothole/damage large or deep enough to be a serious hazard?

YES

Is it affecting vehicle movement or blocking traffic?

NO

🏠

☰

⊕

🔍

Track

👤

Track

Citizens

No account, no password.

Citizen

Employee

Research

Report a problem with just your email. We send a one-time code to confirm it's you, and give you a tracking ID to follow the fix.

File a complaint

Track a complaint

🏠

☰

⊕

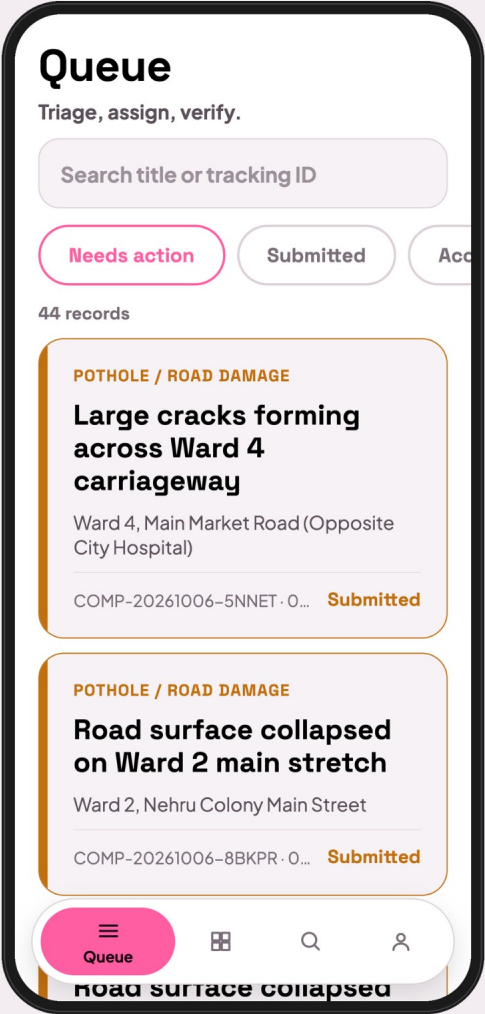
🔍

👤

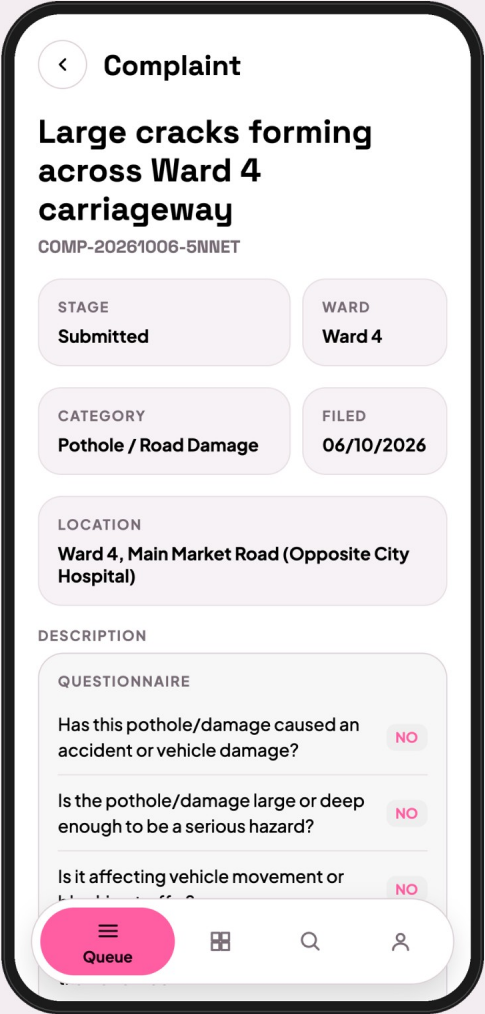
Account

Account

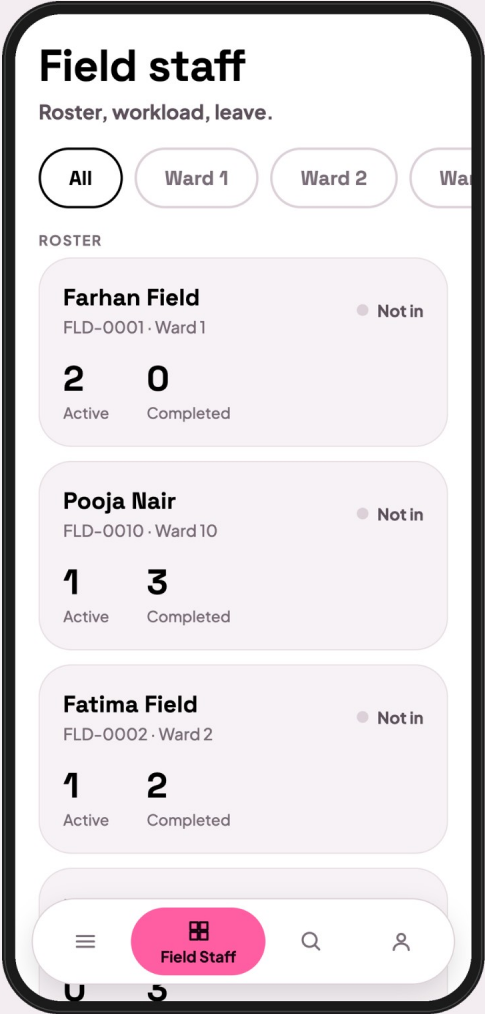
SUPERVISOR



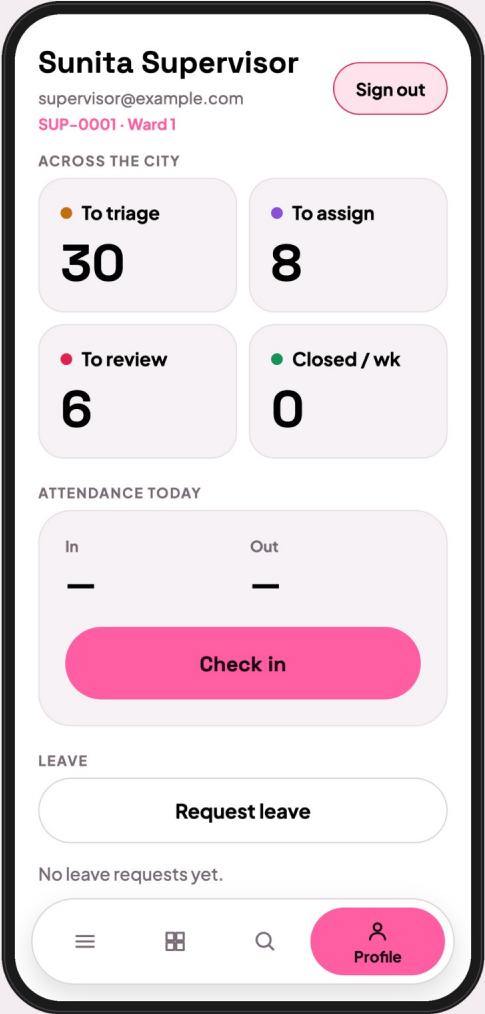
Queue



Triage

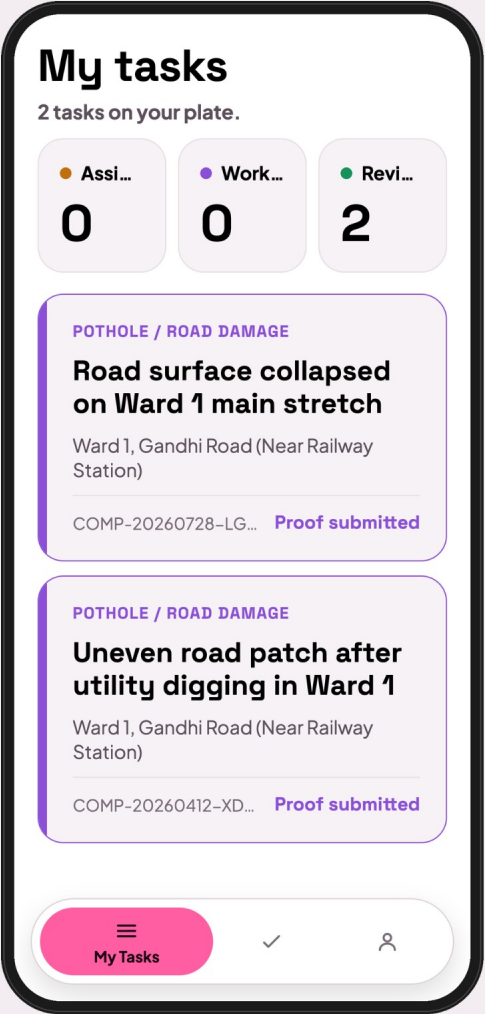


Field staff

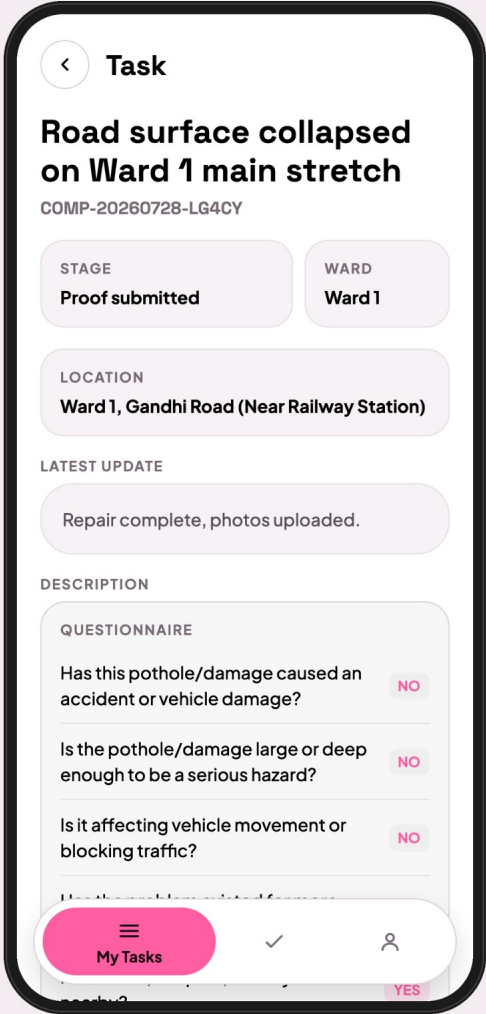


Profile

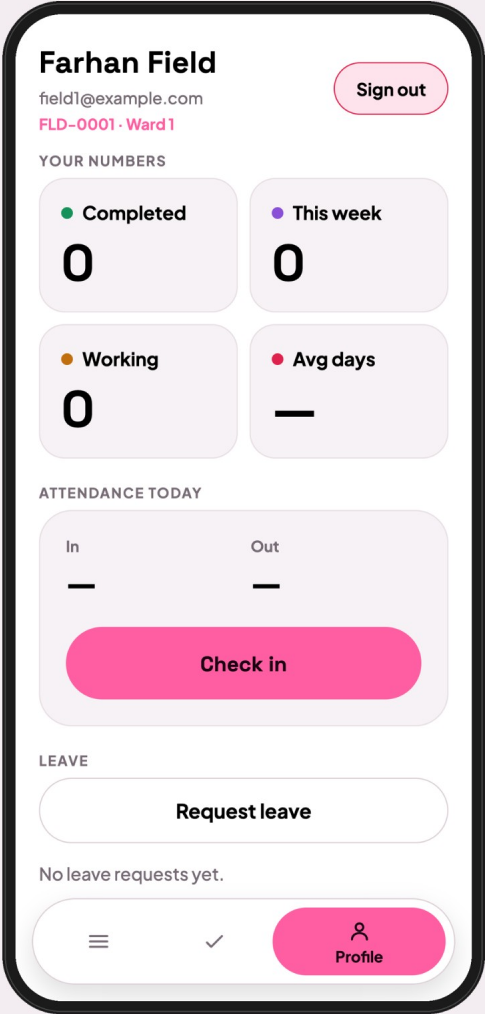
FIELD WORKER



Tasks

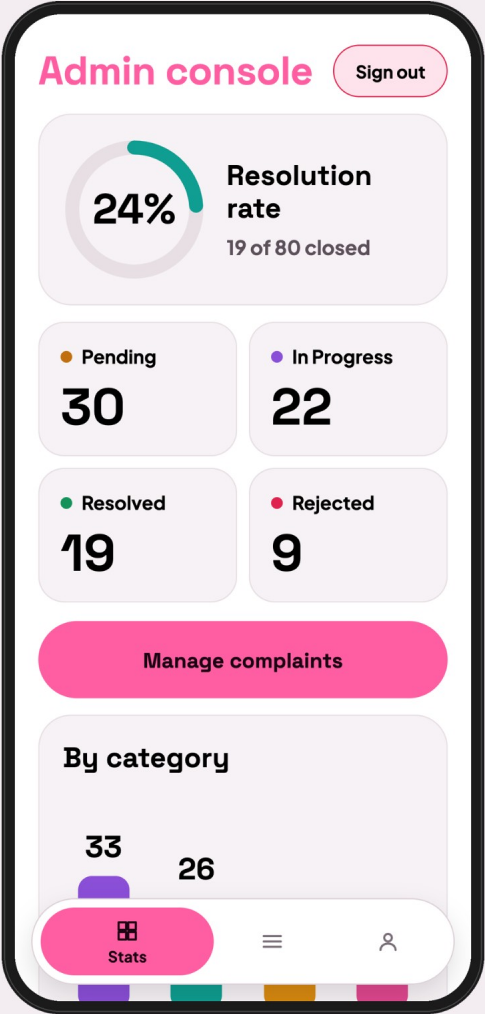


Task and proof

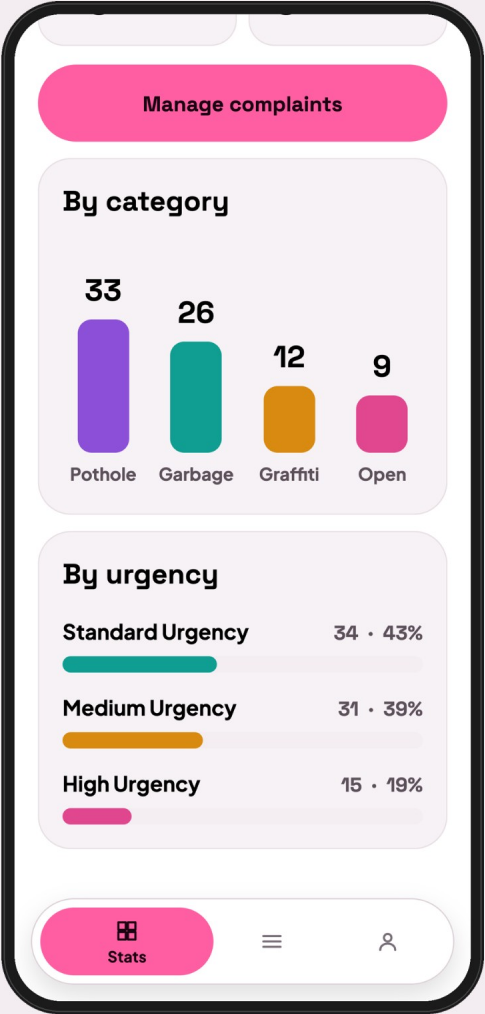


Profile

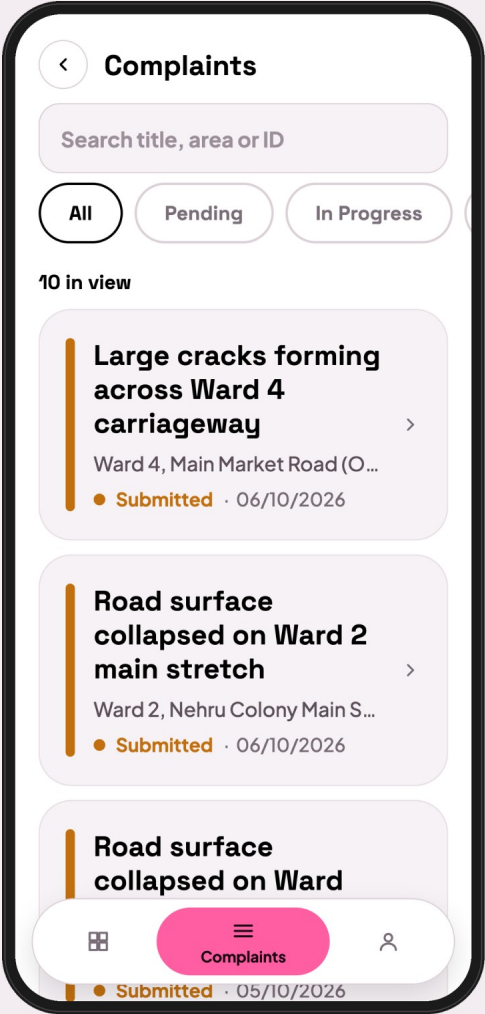
ADMINISTRATOR



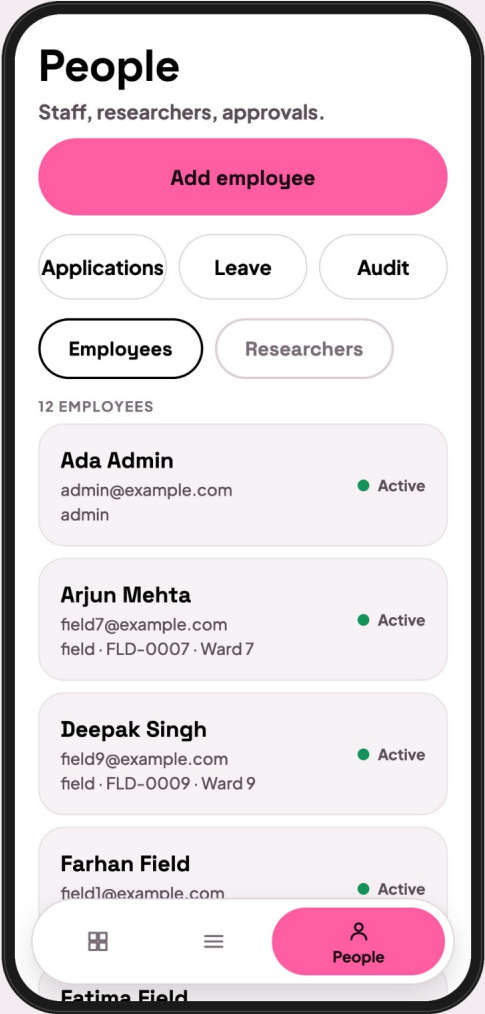
Console



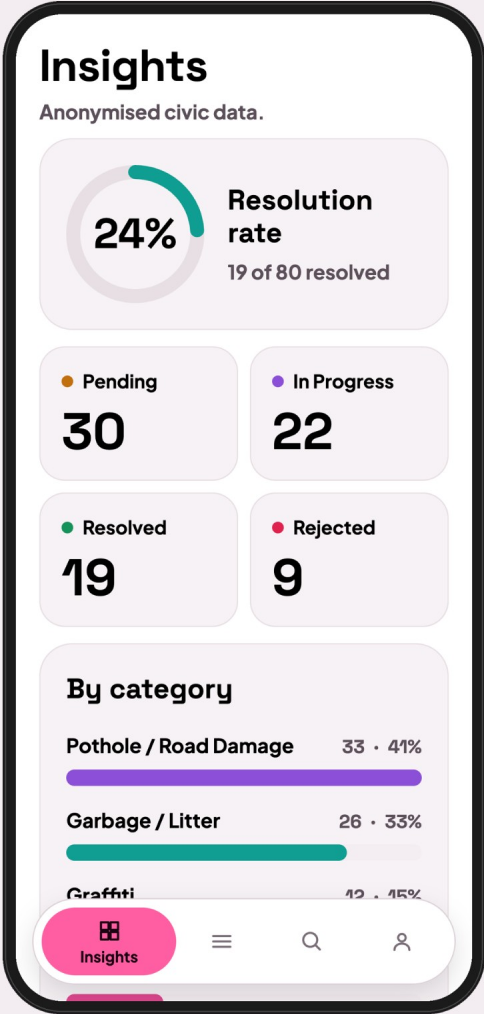
Analytics



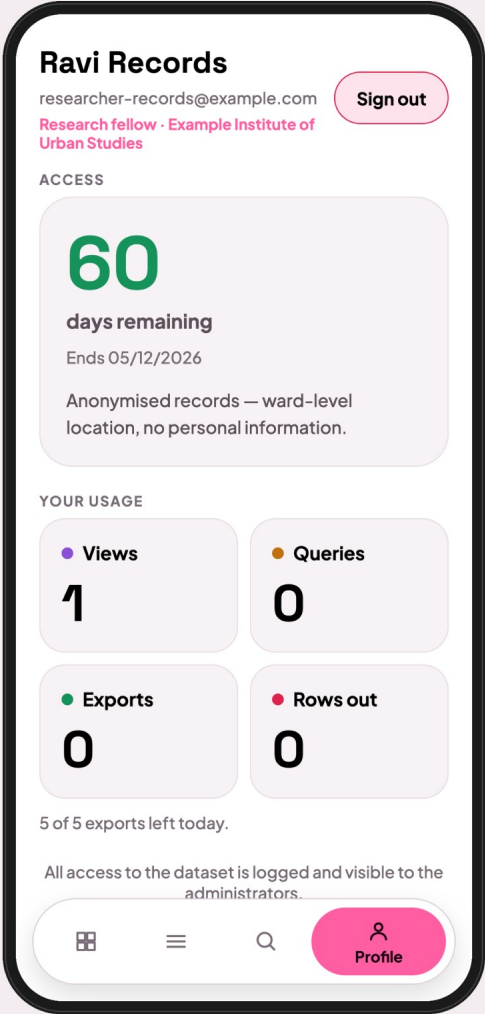
Complaints



People



Insights

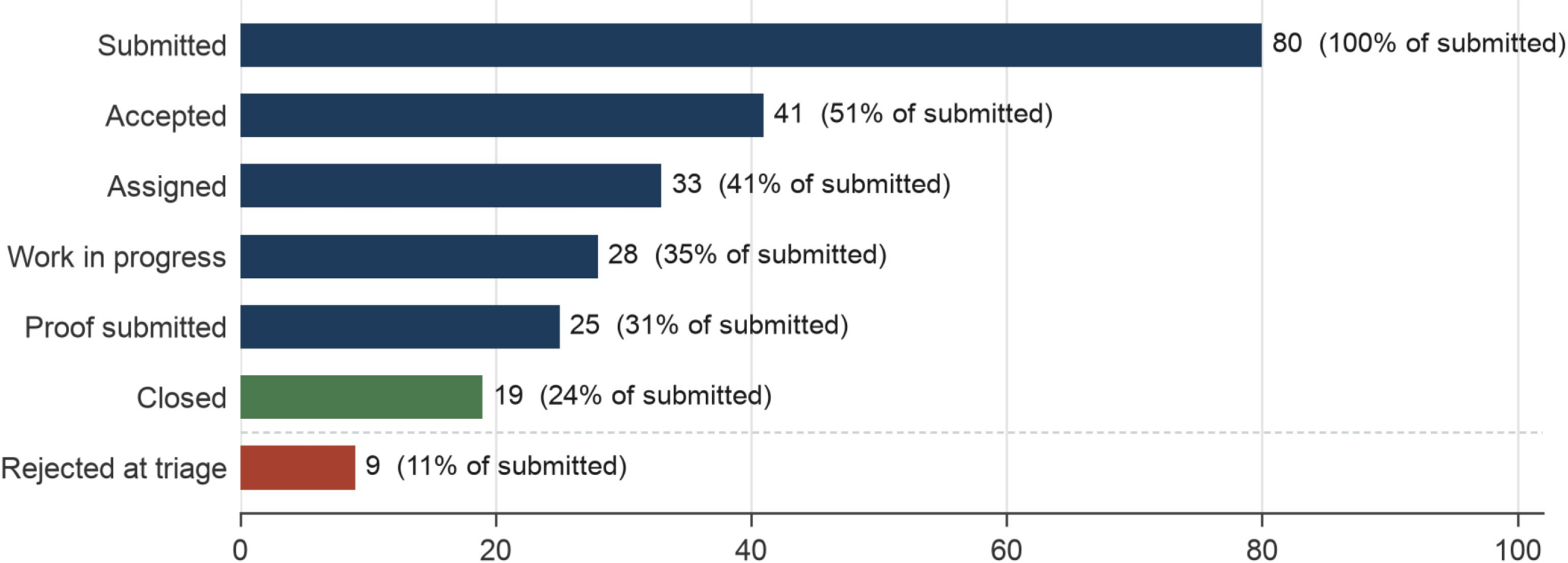


Access and usage

07

Results

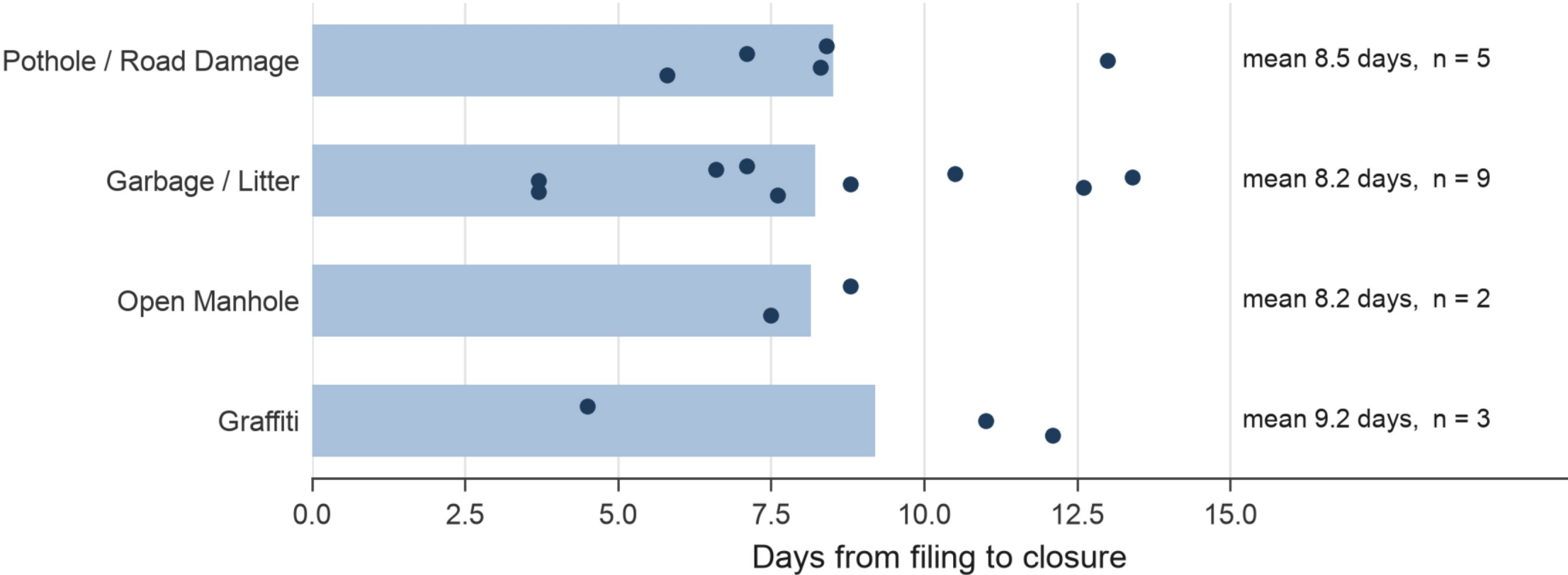
WORKFLOW FUNNEL · DEMO DATA



Complaints that reached the stage

Demo dataset: 80 seeded complaints over 12 months (backend/scripts/seedComplaints.js). A complaint counts once for every stage it has reached.

RESOLUTION TIME · DEMO DATA



Demo dataset (seeded). Bars show the mean; dots show individual closed complaints.

UrbanFix

Report · Validate · Act · Prove · Learn